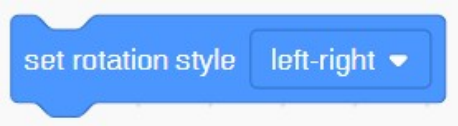
Day-2

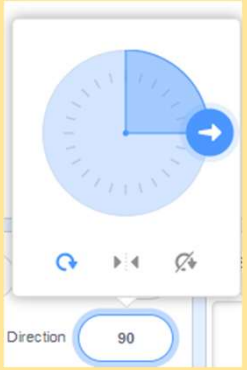
- Digital Artist – Tiffany Newtons
- tdnewton11@yahoo.com
- Scratch – Time to Fly!

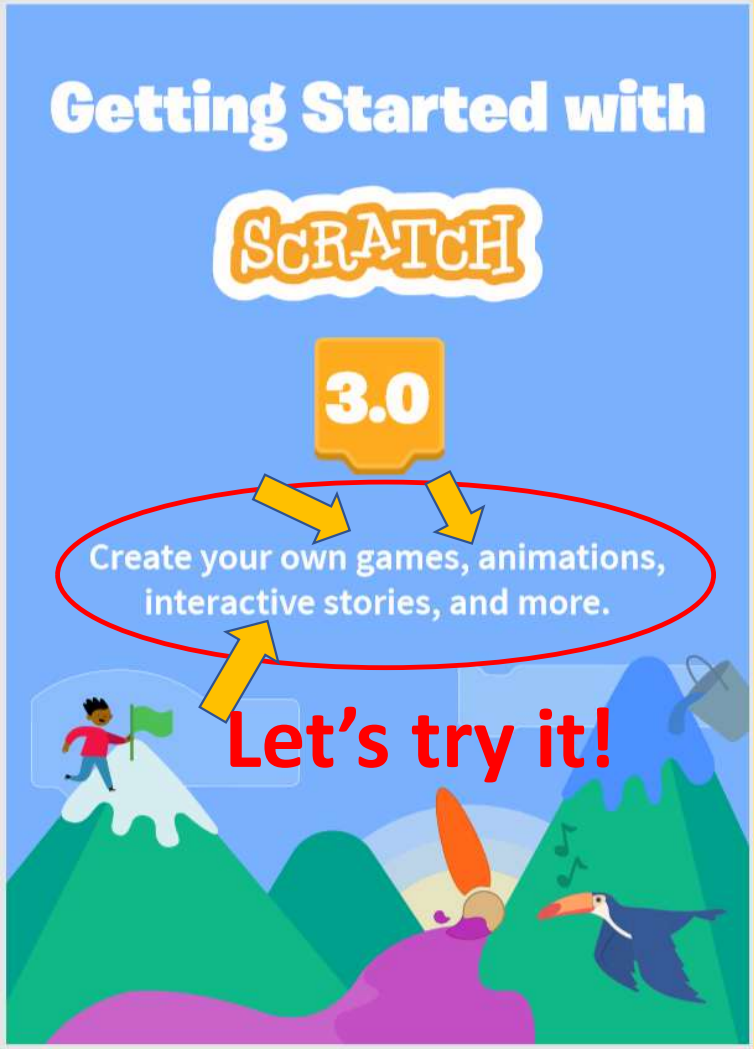
- STEM (Science Technology Engineering Mathletics)
- STEAM (Science Technology Engineering Art Mathletics)
- Some setbacks and solutions …

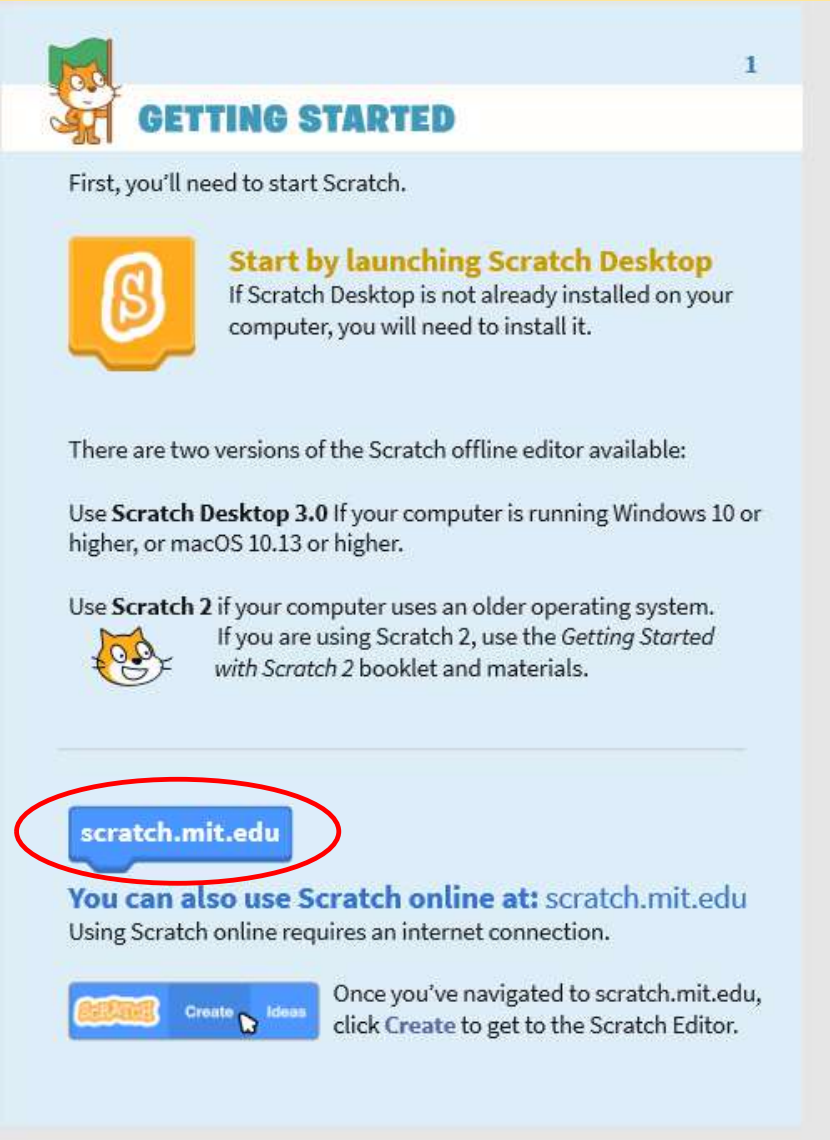
Day-2

- Some setbacks and solutions
  - Rotate 180 degrees (Avery facing right) vs. Flipping









1

GETTING STARTED

First, you'll need to start Scratch.

Start by launching Scratch Desktop
If Scratch Desktop is not already installed on your computer, you will need to install it.

There are two versions of the Scratch offline editor available:

Use **Scratch Desktop 3.0** If your computer is running Windows 10 or higher, or macOS 10.13 or higher.

Use **Scratch 2** if your computer uses an older operating system. If you are using Scratch 2, use the *Getting Started with Scratch 2* booklet and materials.

scratch.mit.edu

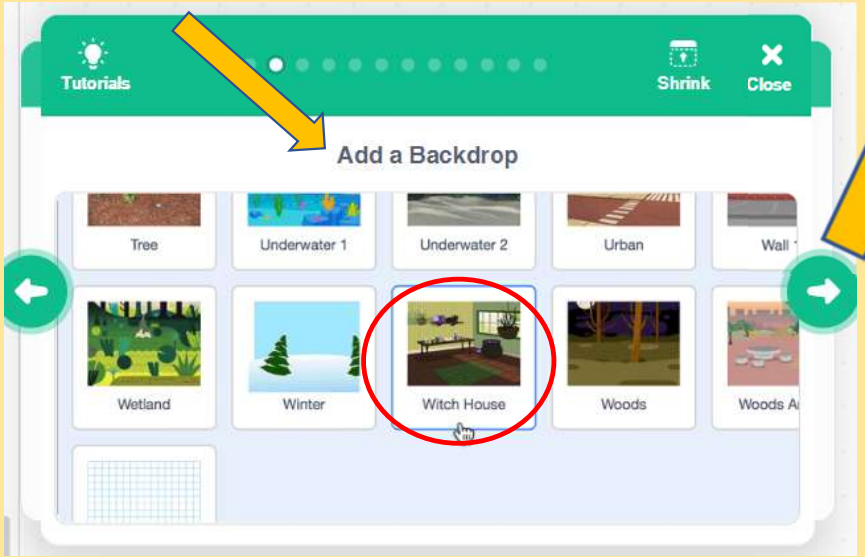
You can also use Scratch online at: scratch.mit.edu
Using Scratch online requires an internet connection.

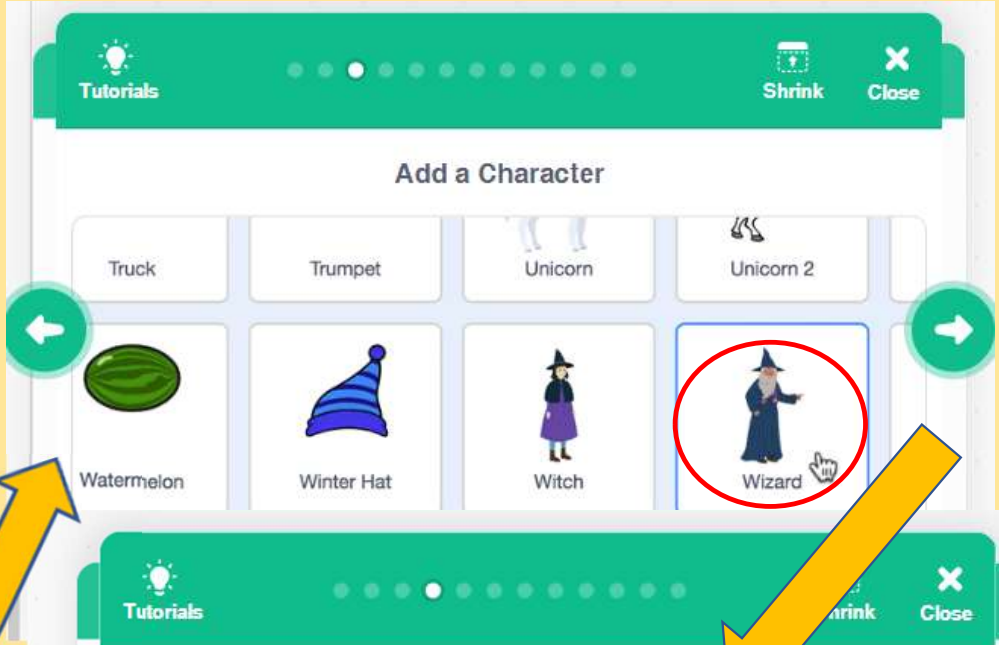
Create Ideas

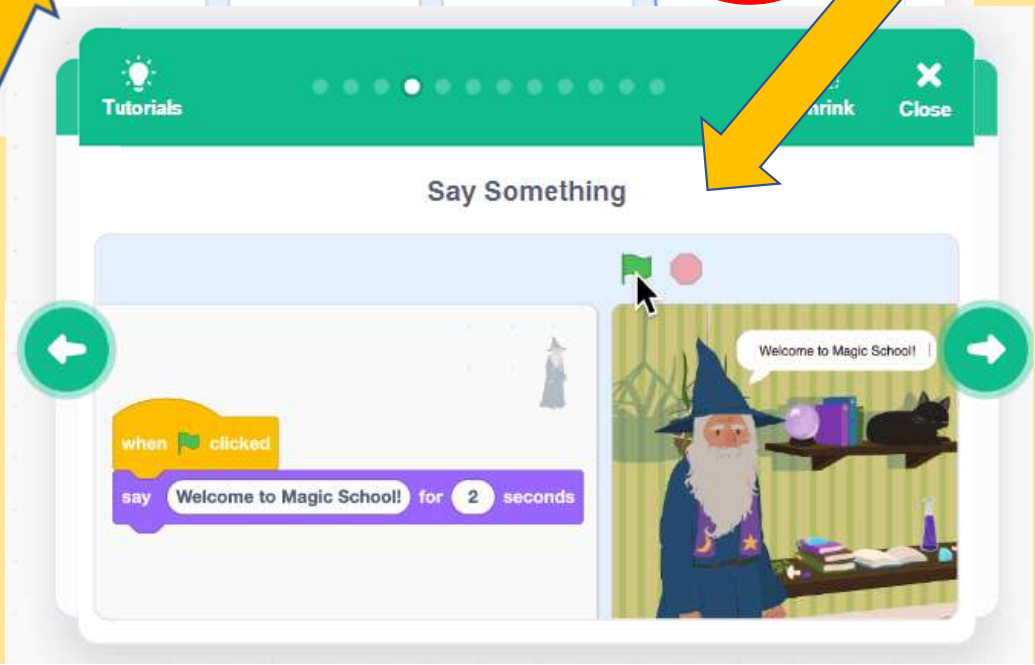
Once you've navigated to scratch.mit.edu, click **Create** to get to the Scratch Editor.

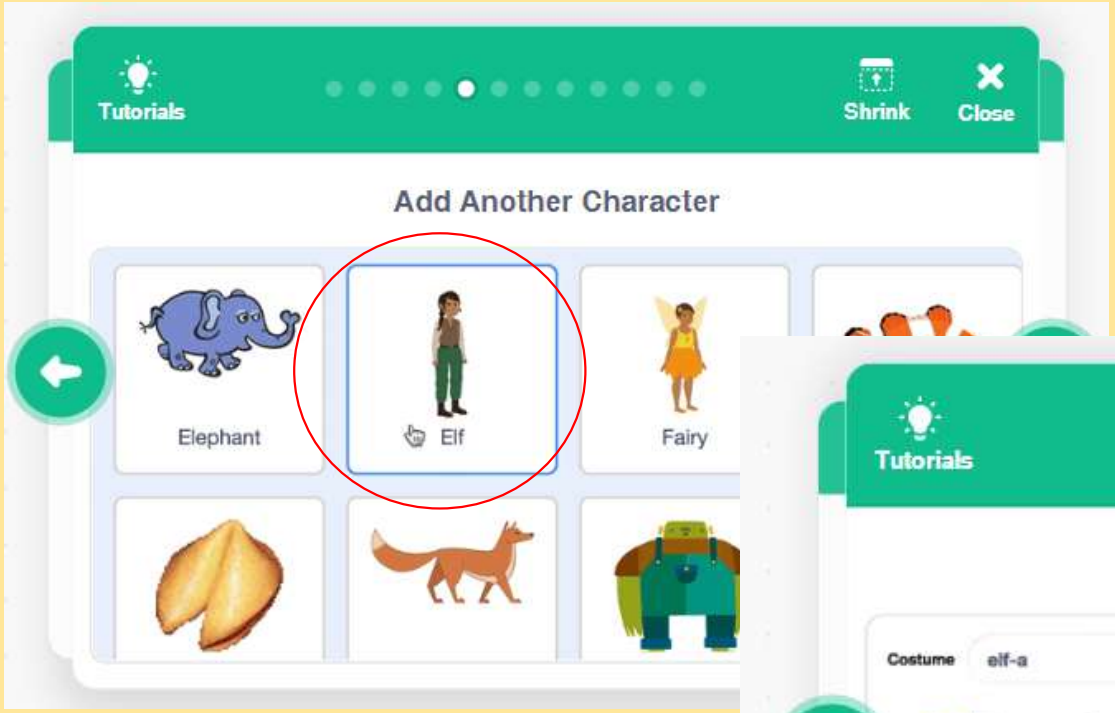
file:///H:/temp/IoT/Getting-Started-With-Scratch-3.0.pdf

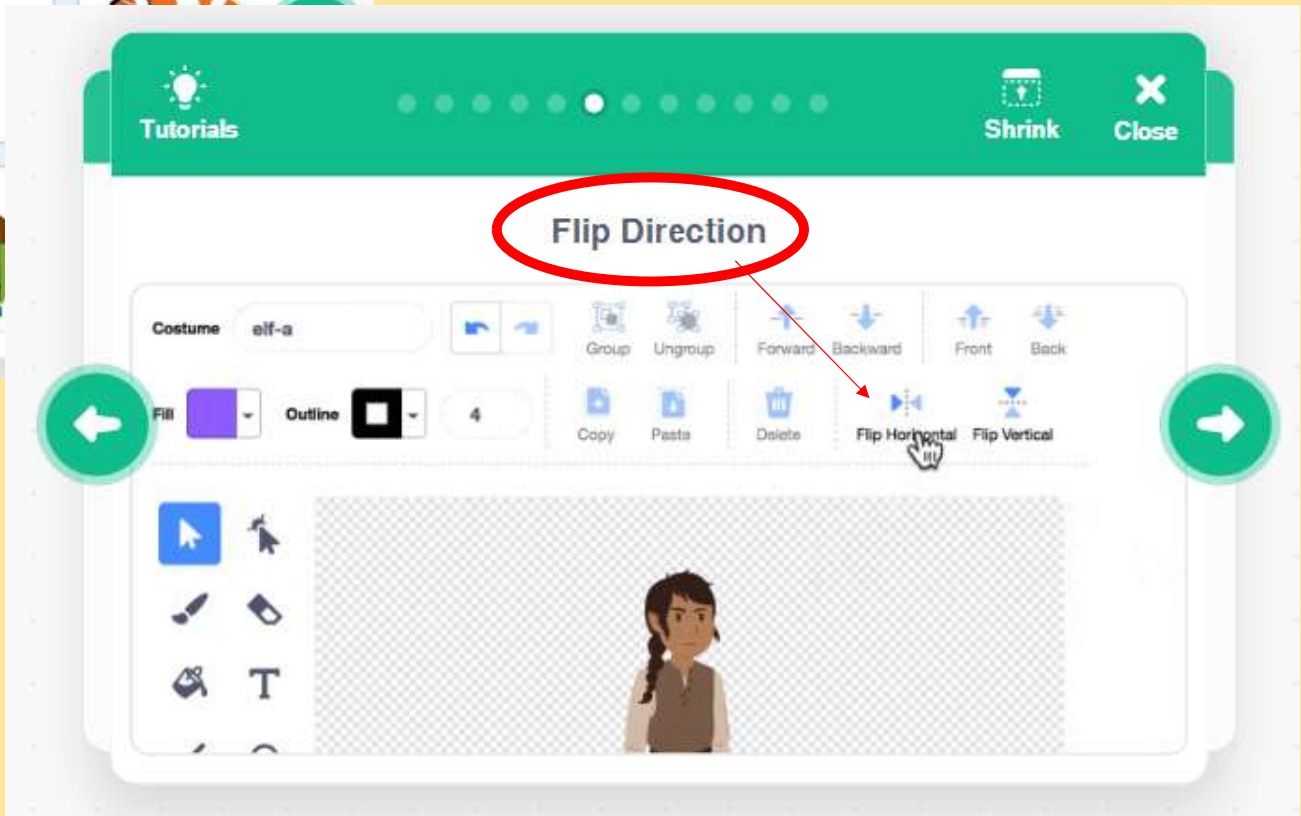
***** Reminder: Watch the video first!***

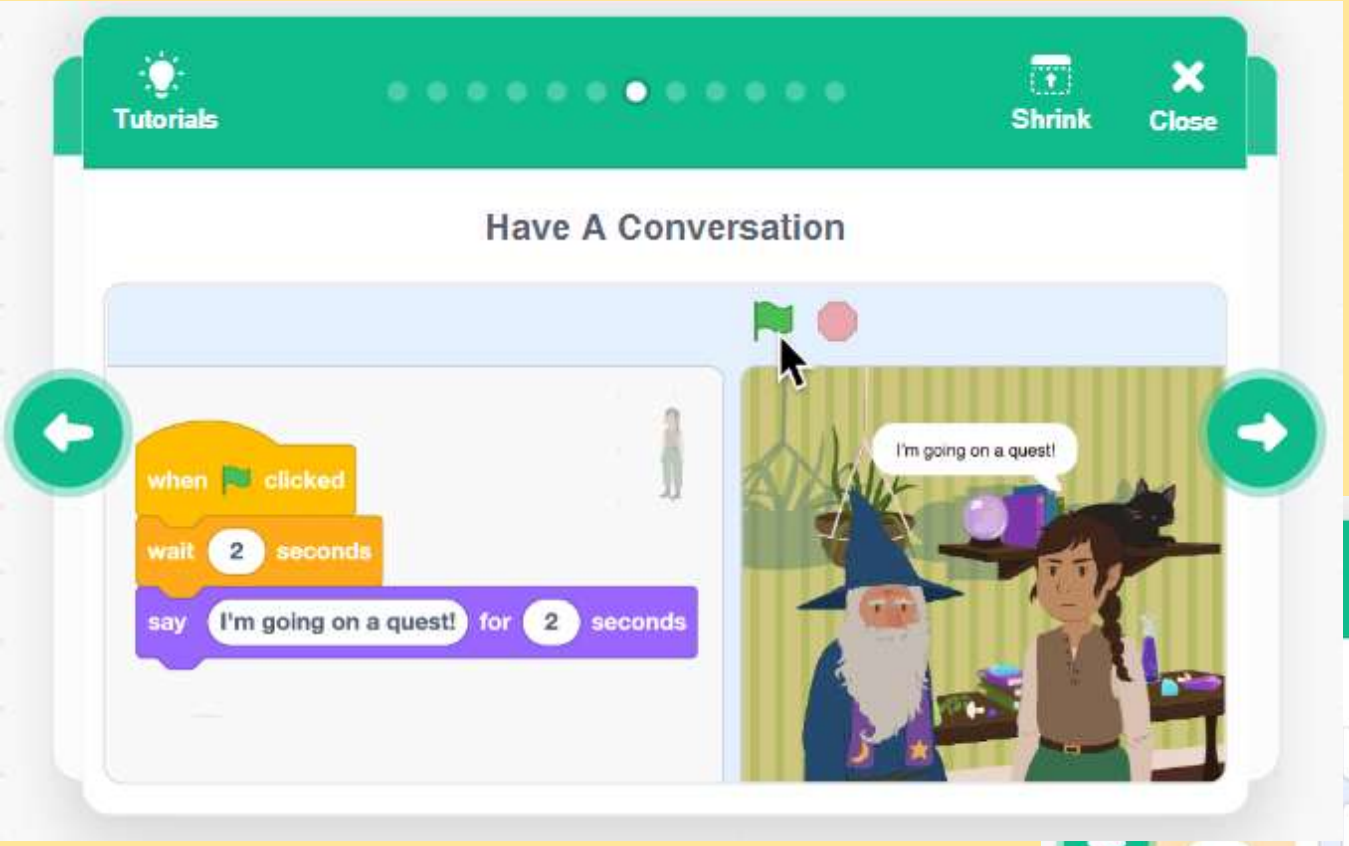


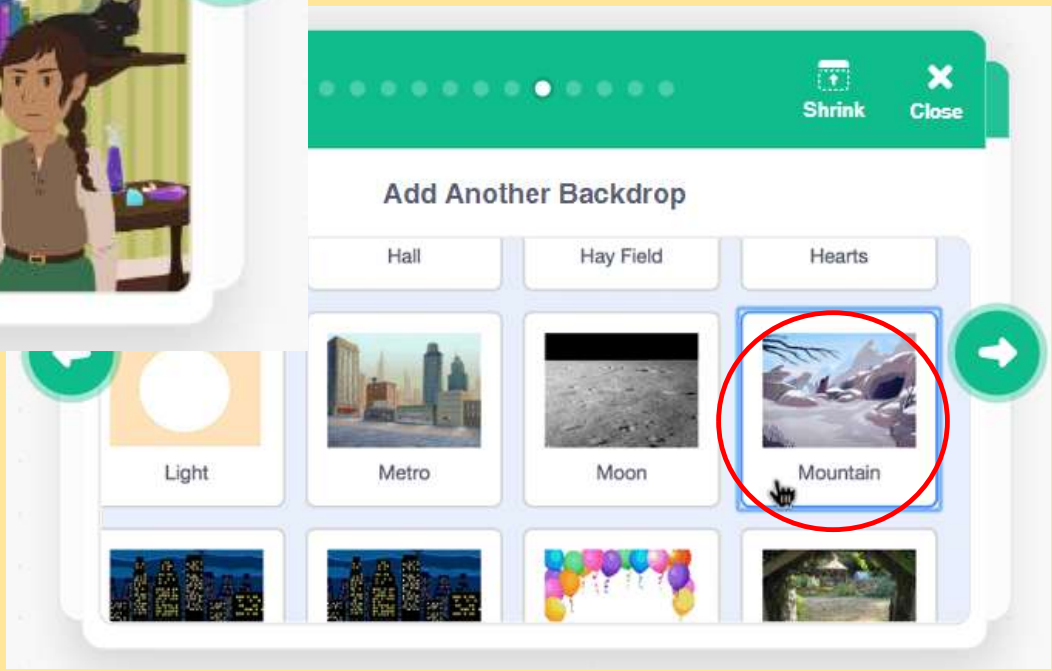


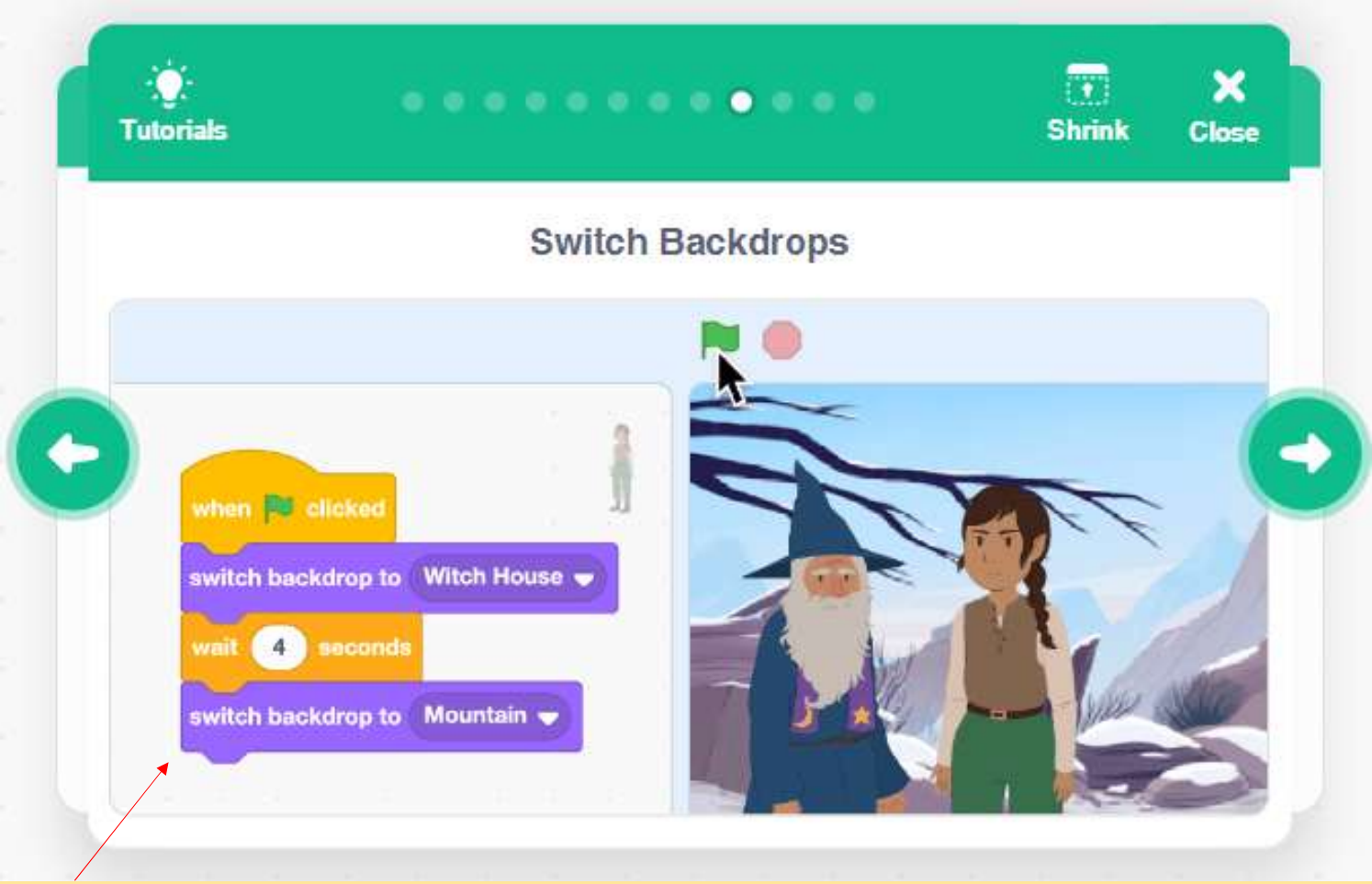


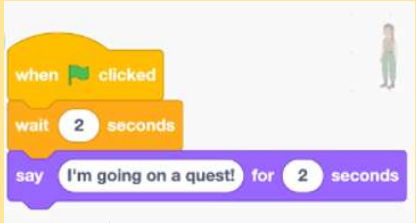

Tutorials
Shrink
Close
Add Another Character
Elephant
Elf
Fairy


Tutorials
Shrink
Close
Flip Direction
Costume
elf-a
Group
Ungroup
Forward
Backward
Front
Back
Fill
Outline
4
Copy
Paste
Delete
Flip Horizontal
Flip Vertical


Tutorials
Shrink
Close
Have A Conversation
when clicked
wait 2 seconds
say I'm going on a quest! for 2 seconds
I'm going on a quest!

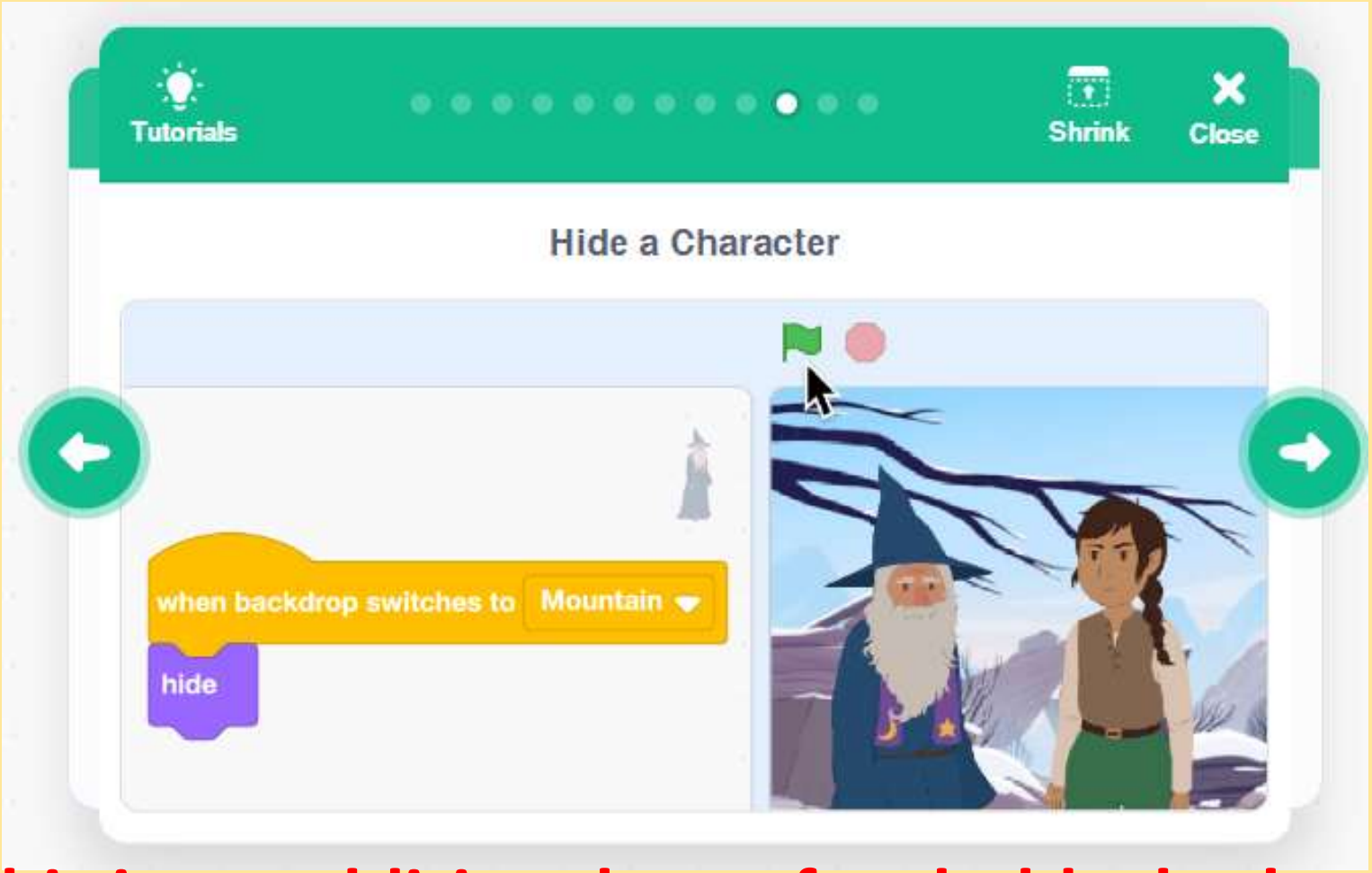

Shrink
Close
Add Another Backdrop
Hall
Hay Field
Hearts
Light
Metro
Moon
Mountain

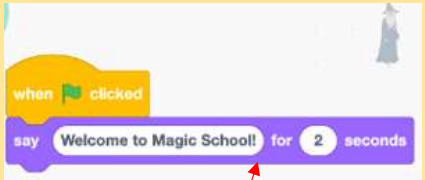




This is an additional set of code-blocks, beside this set.

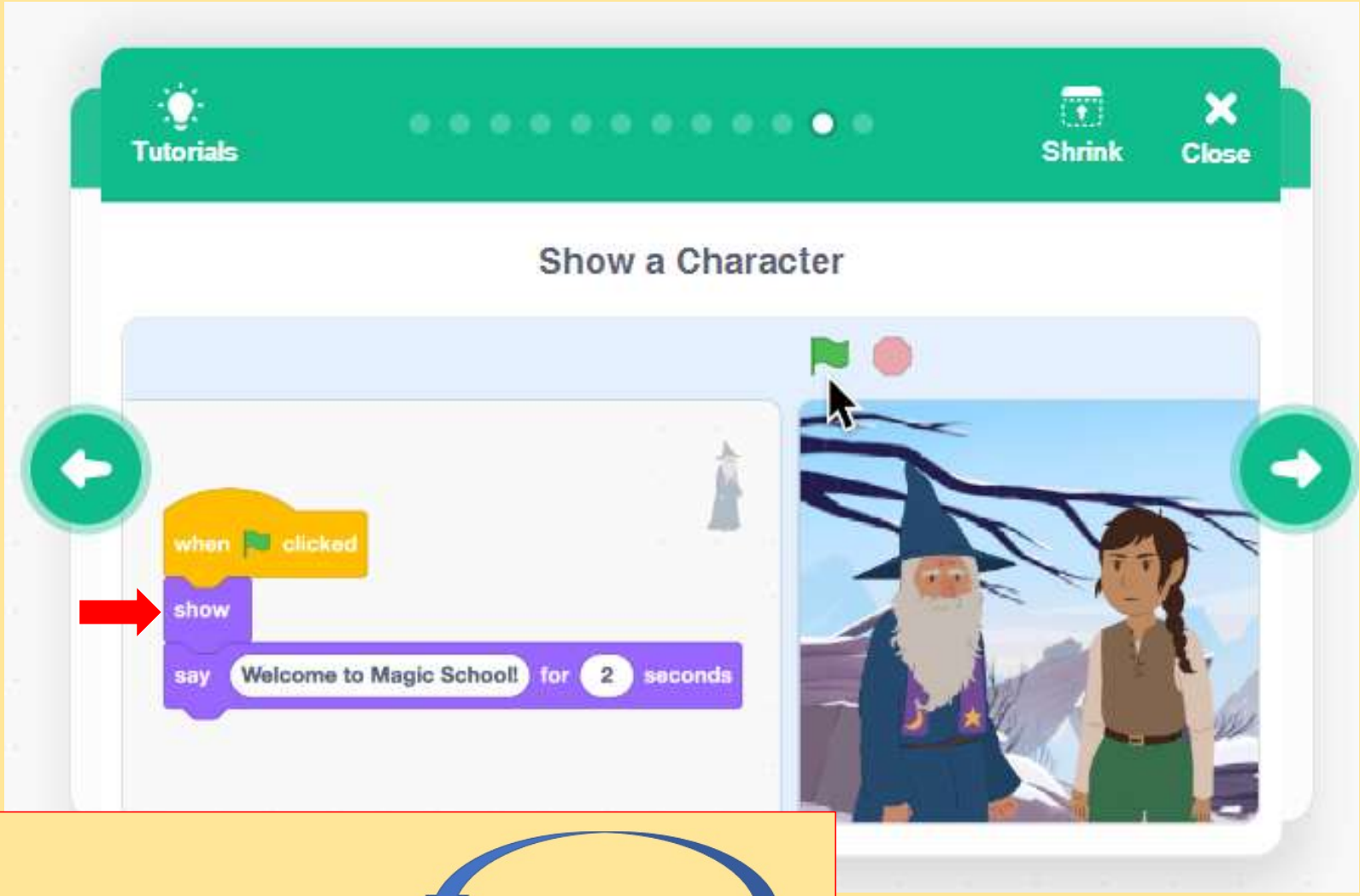
Why wait for 4 seconds? [parallel processing/threading]





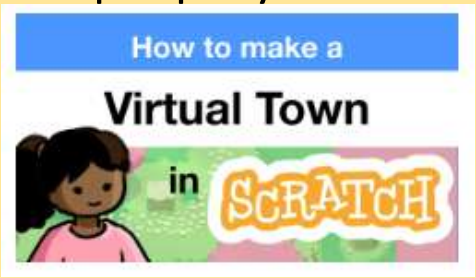
This is an additional set of code-blocks, beside this set.

Event [event programming]



Software Development Life Cycle: Design, Coding, Testing, Debugging

- Any Questions?
- Applications?
  - Tell people your town!



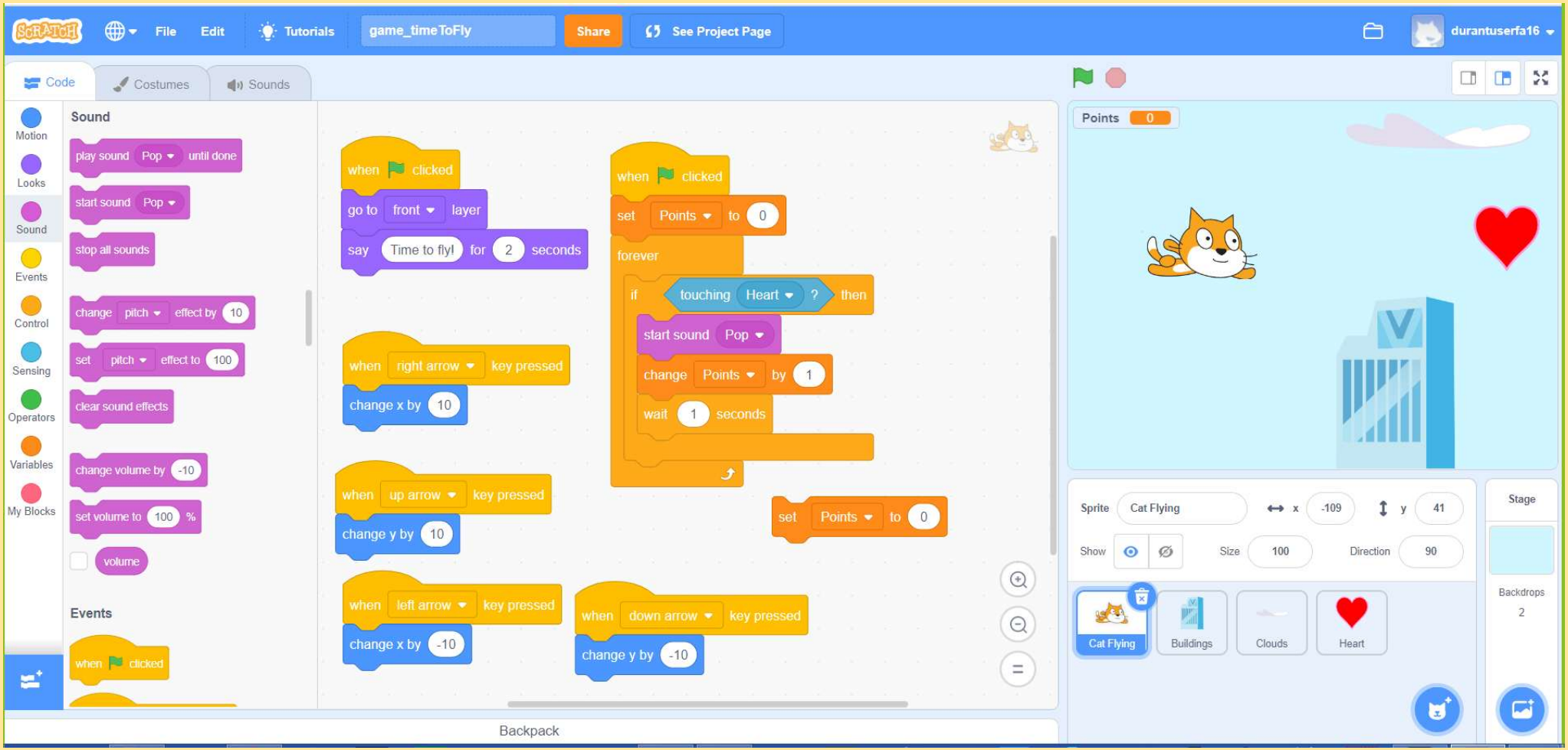
https://youtu.be/eekrc3Xs9Z0 [8:51]

  - Design your own Character (Sprite, Avatar)

https://youtu.be/-3oCdNIeU_8 [7:35]

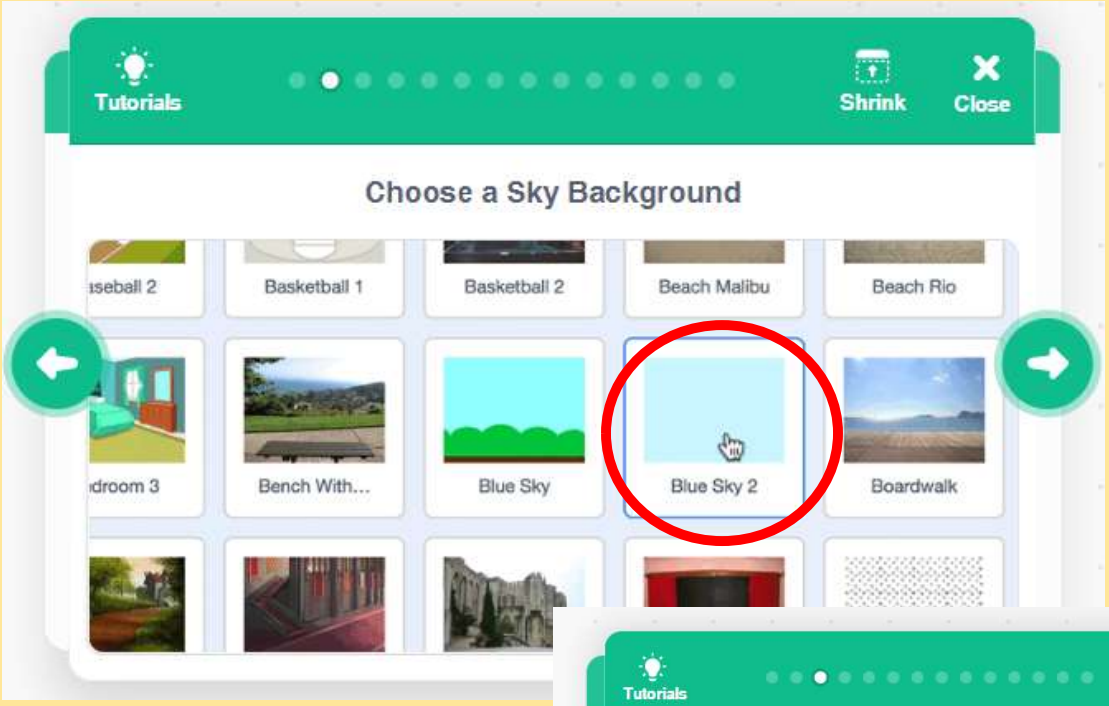
Game: Time to fly! (simple)

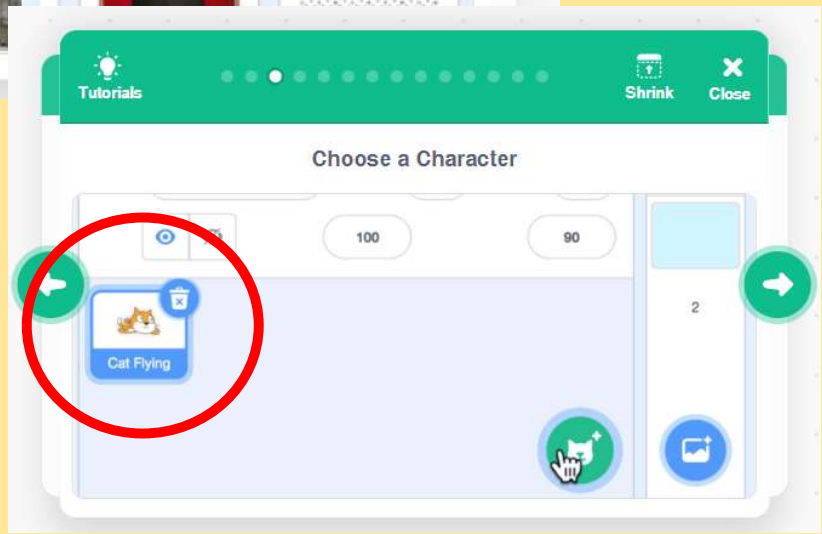
Put them all together! (Variable, Looping, Selection, Events, …)



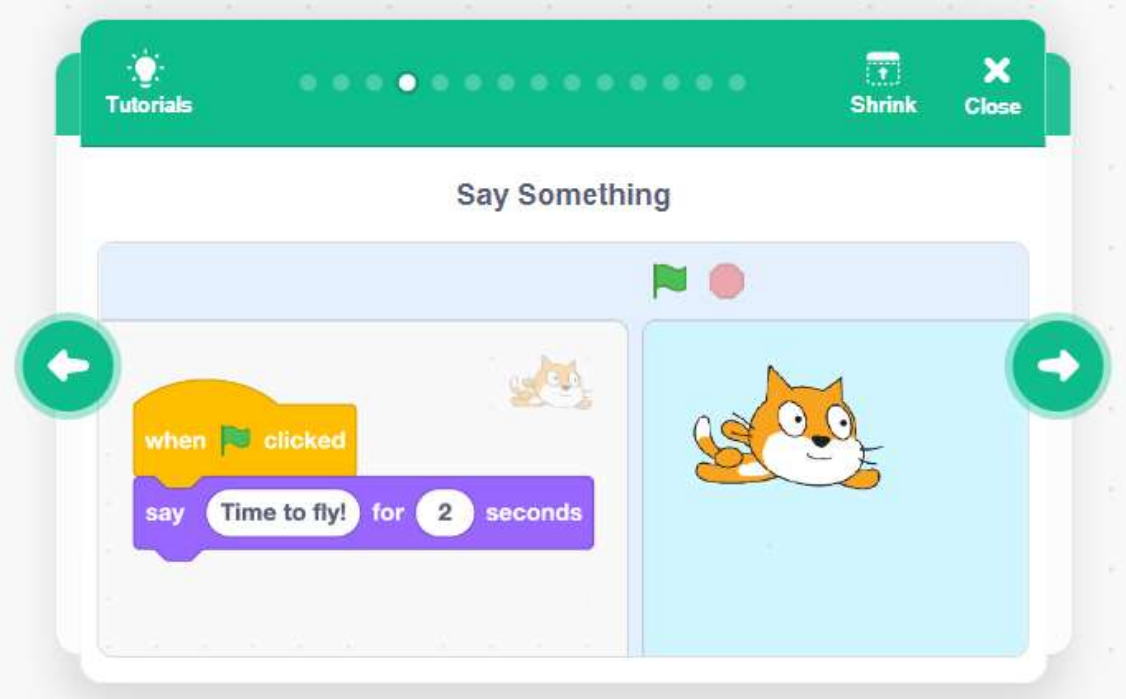
https://youtu.be/uBTFqUr7KPA

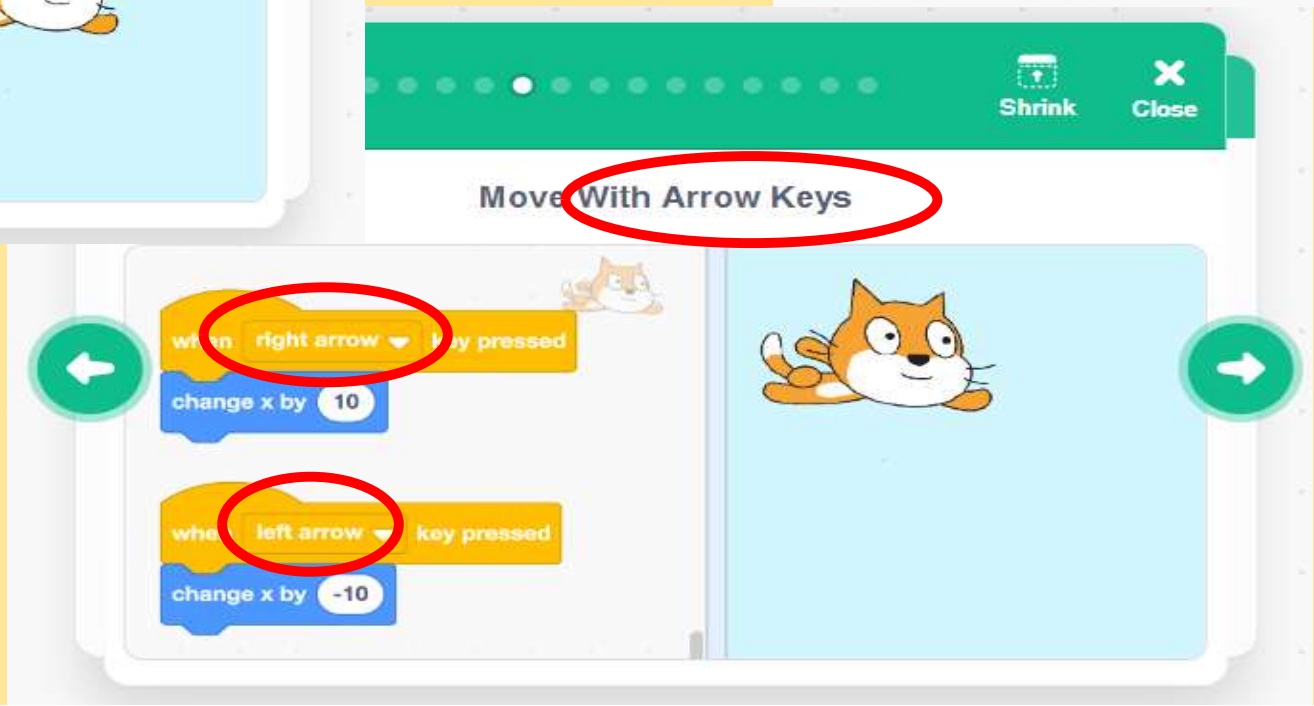
***** Reminder: Watch the video first!***

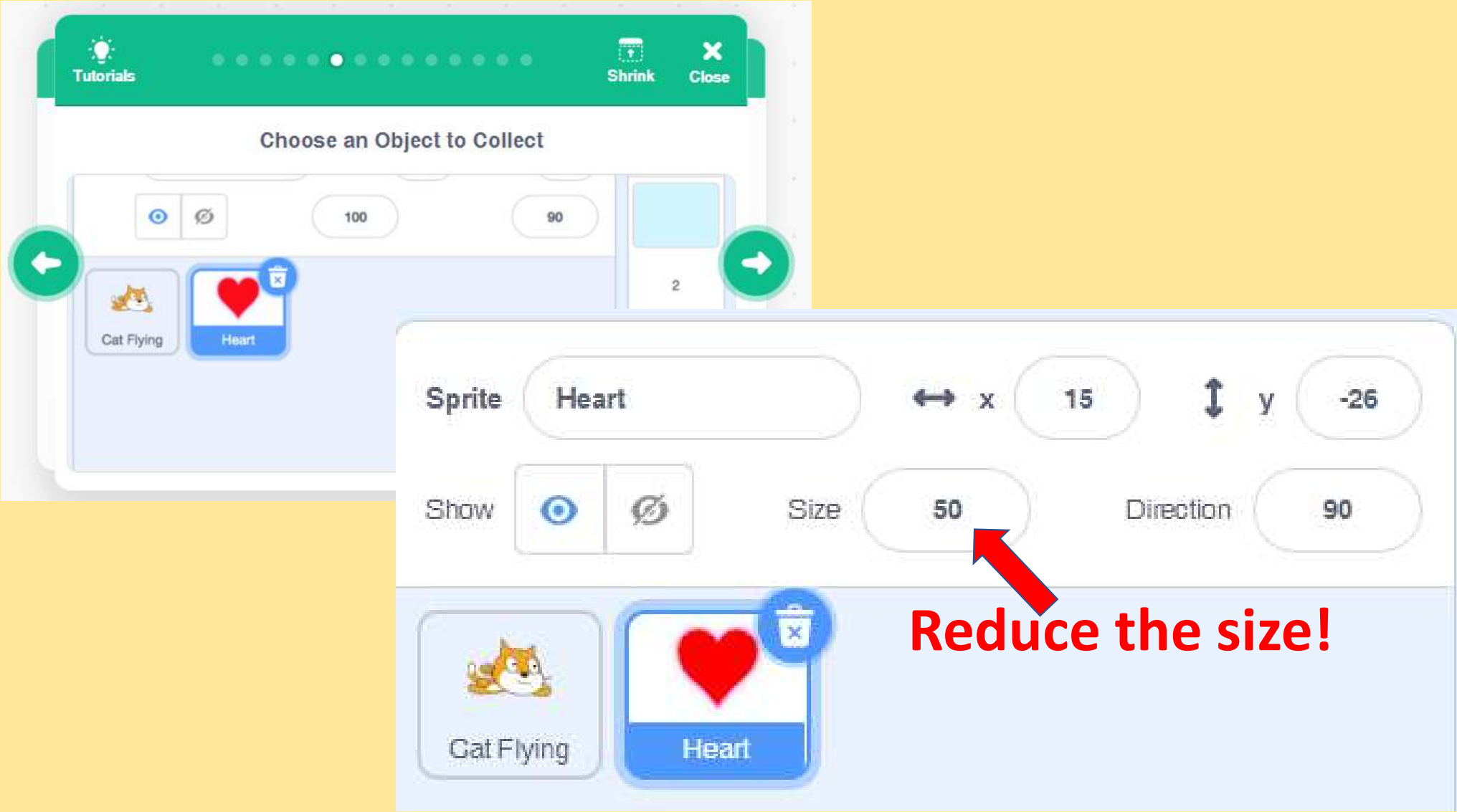


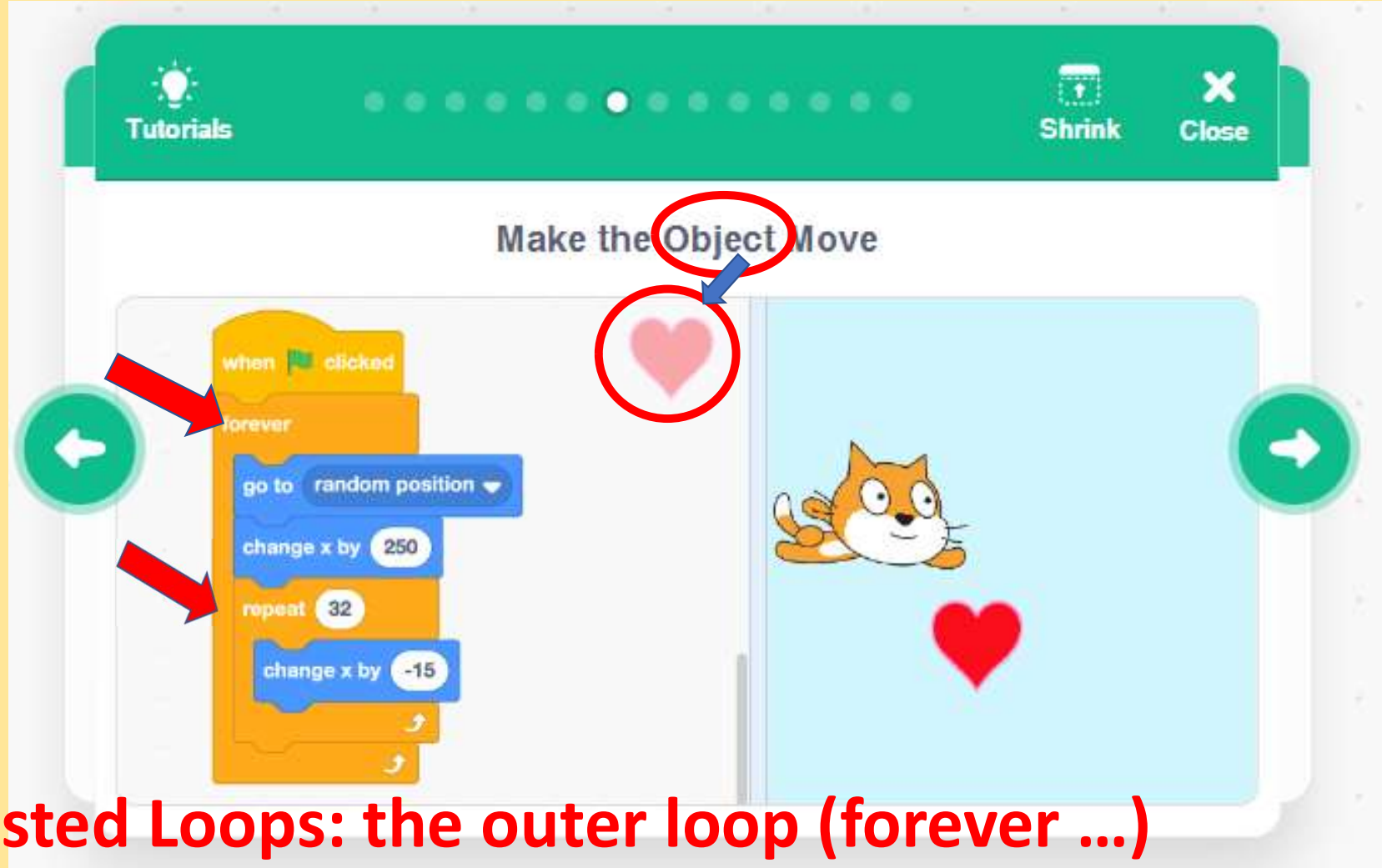


How about Up and Down?






Tutorials
Shrink
Close
Choose an Object to Collect
100
90
2
Cat Flying
Heart
Sprite
Heart
x
15
y
-26
Show
Size
50
Direction
90
Cat Flying
Heart
Reduce the size!



Nested Loops: the outer loop (forever …)

and the inner loop (repeat …)

32x-15=-480 Why?

WHAT IS A SPRITE?

Avatar, Actor, or ...

In Scratch, any character or object is called a sprite. Every new project in Scratch starts with the Cat sprite.

Want to choose a different sprite?

Click the New Sprite icon.

Or, hover over the "New Sprite" icon to see more options.

Upload an image from your computer.

Draw your own sprite.

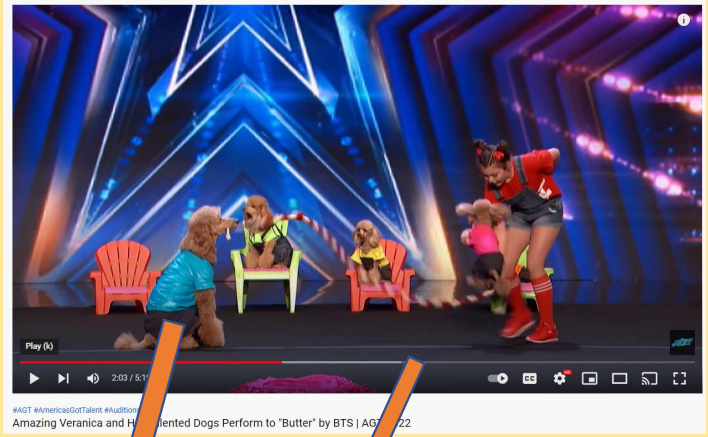
Click for a surprise sprite!

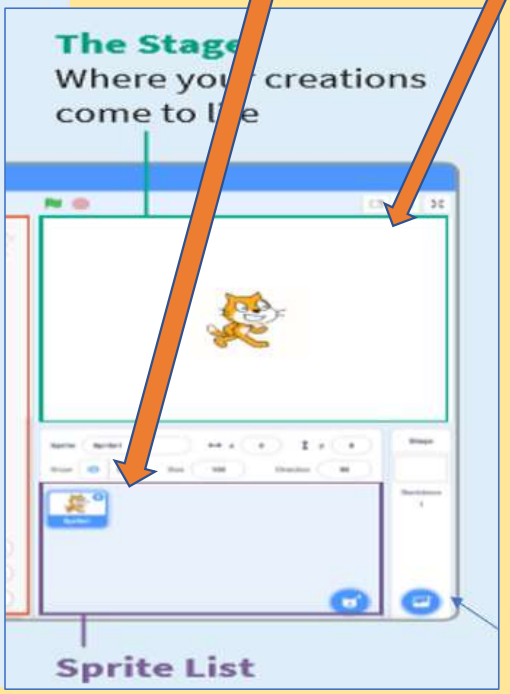
Choose a sprite from the library.

Want to **delete a sprite** from your project?

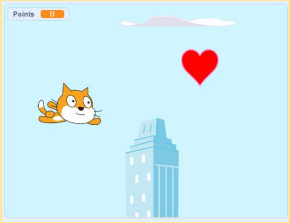
First, select the sprite by clicking on its thumbnail in the Sprite List.

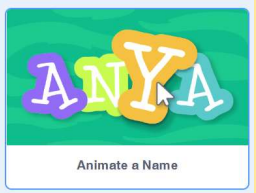
Then, click here to delete the sprite.





Objects:
heart, building, cloud, letters





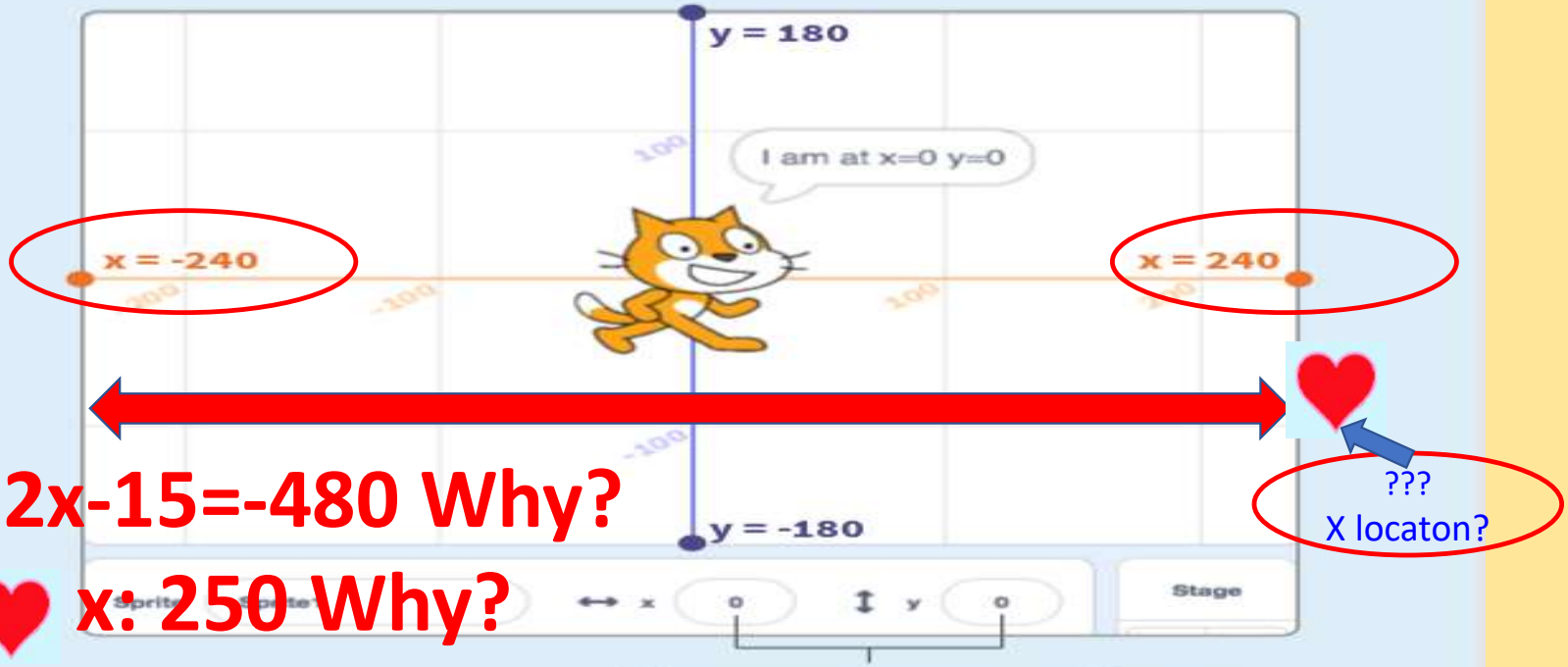
WHERE IS YOUR SPRITE?

Every sprite has an **x** and **y** position on the Stage.

x is the position of the sprite from left-to-right.

y is the position from top-to-bottom.

At the very center of the stage, **x** is 0 and **y** is 0.



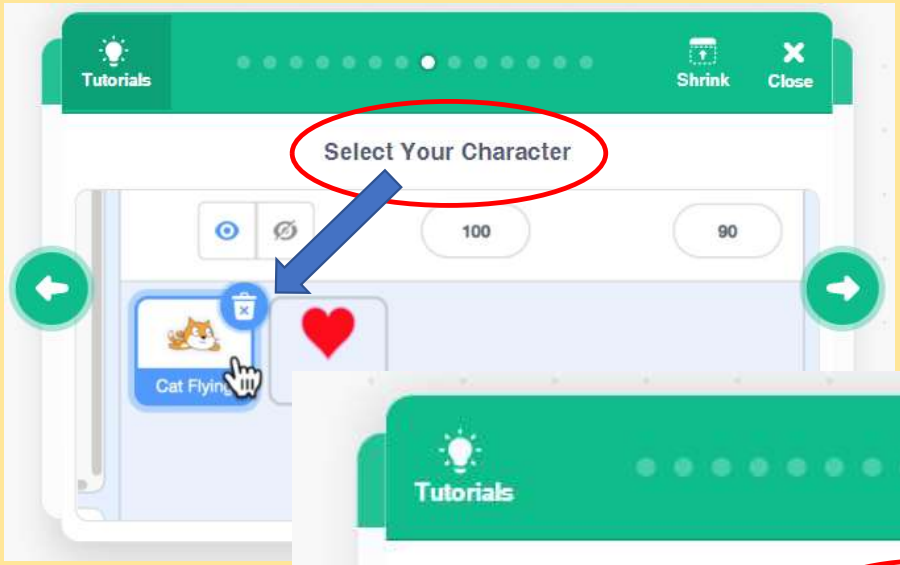
32x-15=-480 Why?

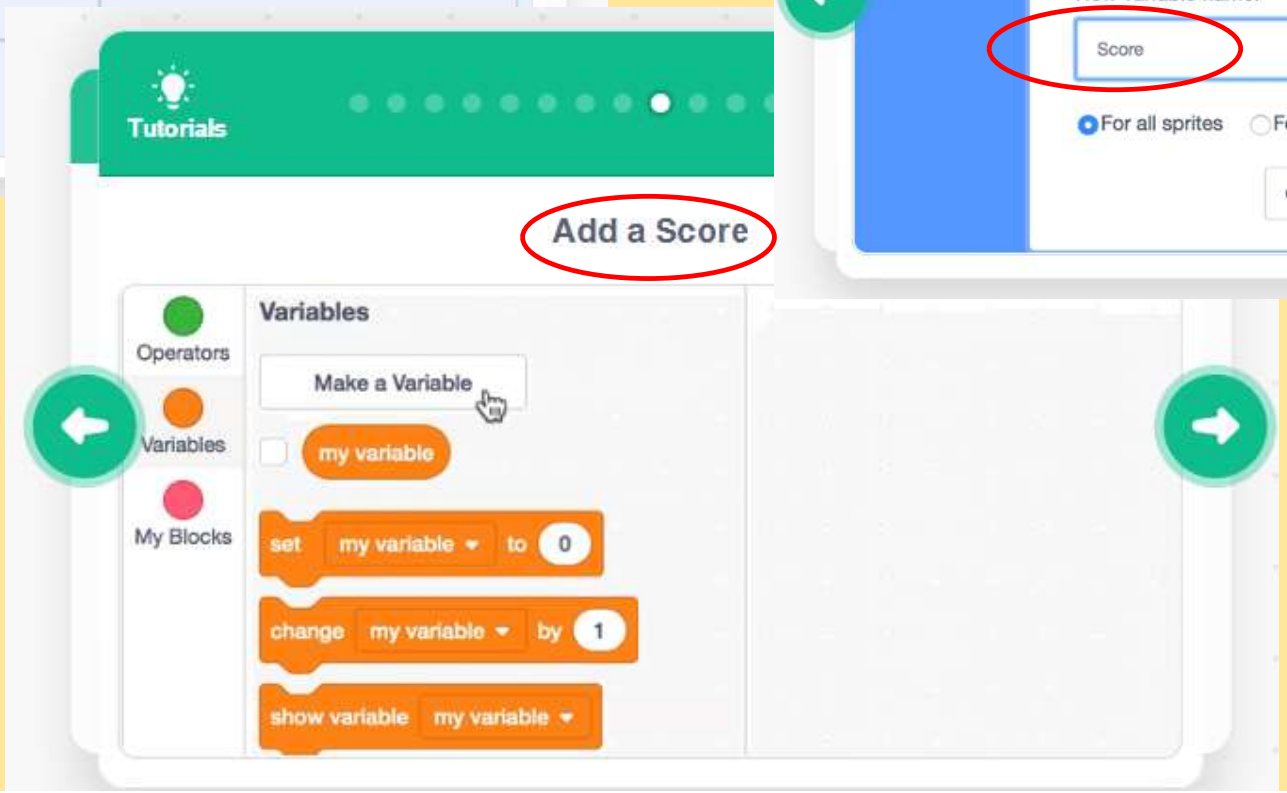
x: 250 Why?

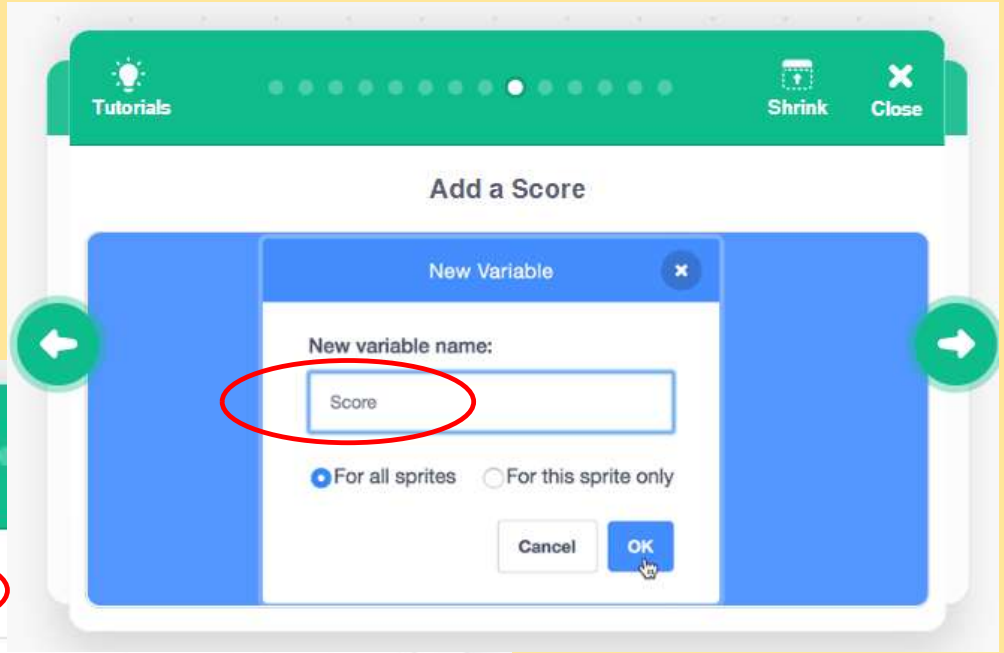
???

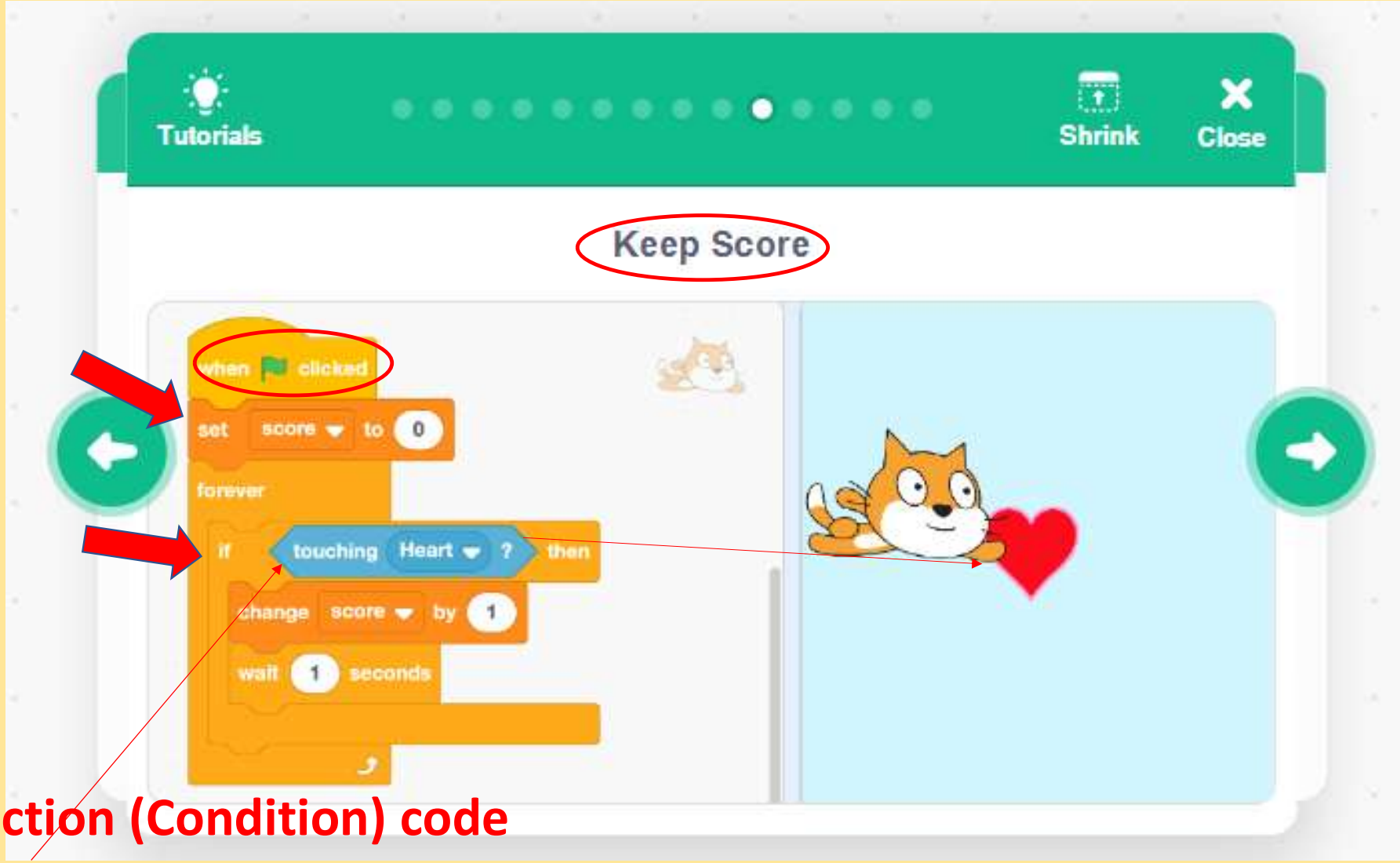
X locaton?

When you move your sprite, you can see its **x** and **y** position change.


Tutorials
Shrink
Close
Select Your Character
100
90
Cat Flying

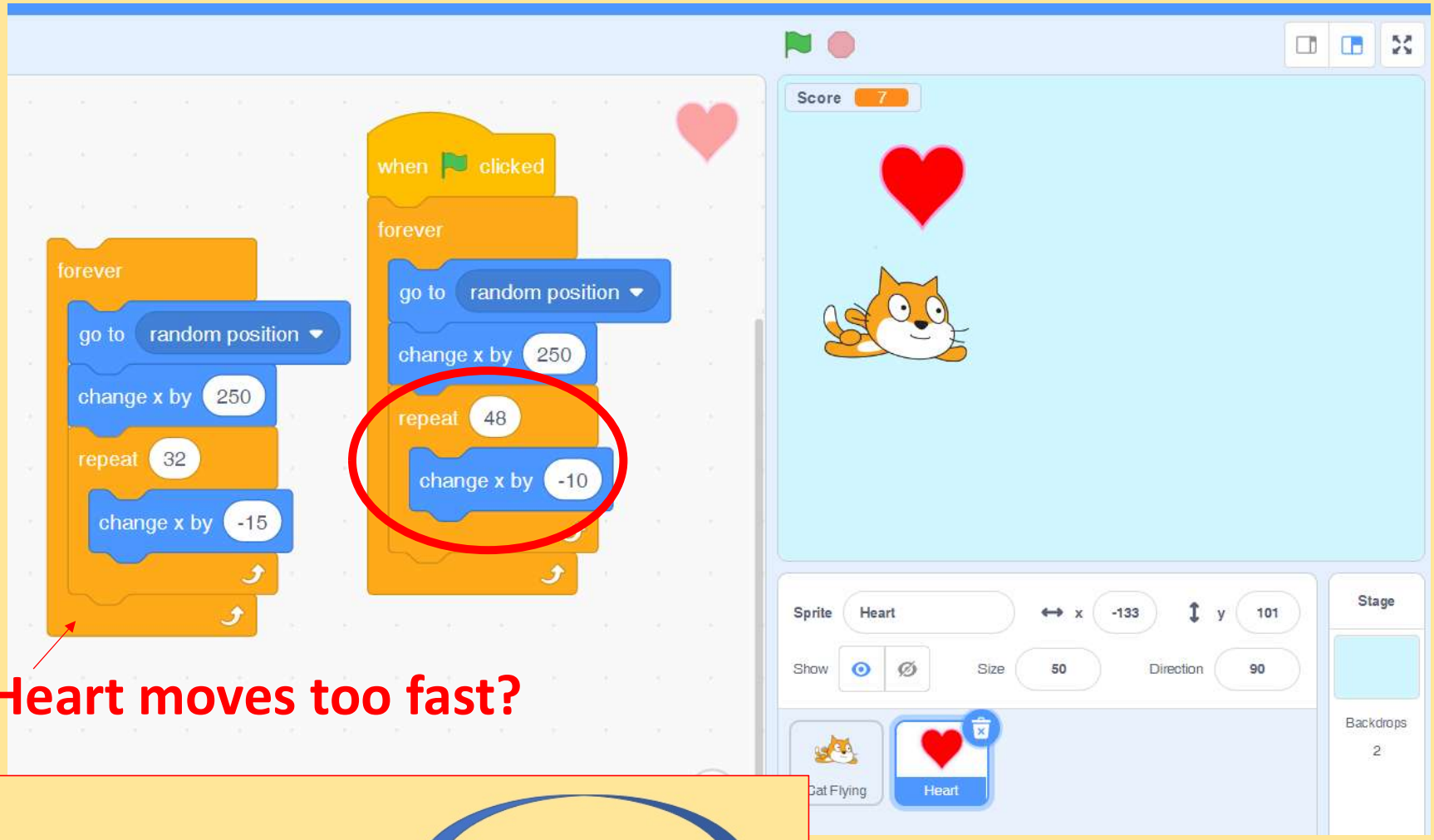

Tutorials
Add a Score
Operators
Variables
My Blocks
Variables
Make a Variable
my variable
set my variable to 0
change my variable by 1
show variable my variable


Tutorials
Shrink
Close
Add a Score
New Variable
New variable name:
Score
For all sprites
For this sprite only
Cancel
OK



Selection (Condition) code

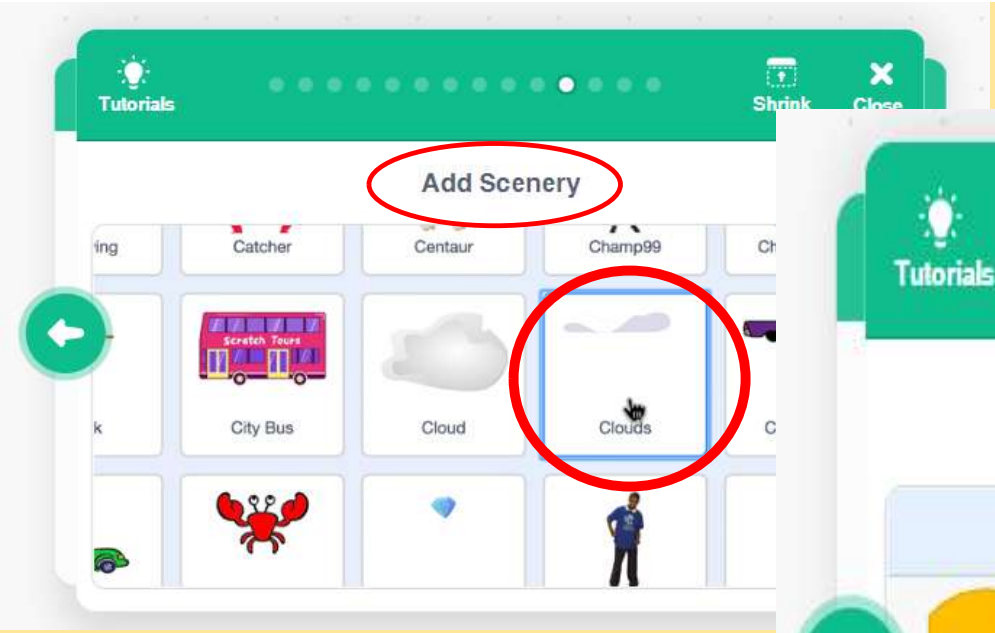
If condition_is_true then …

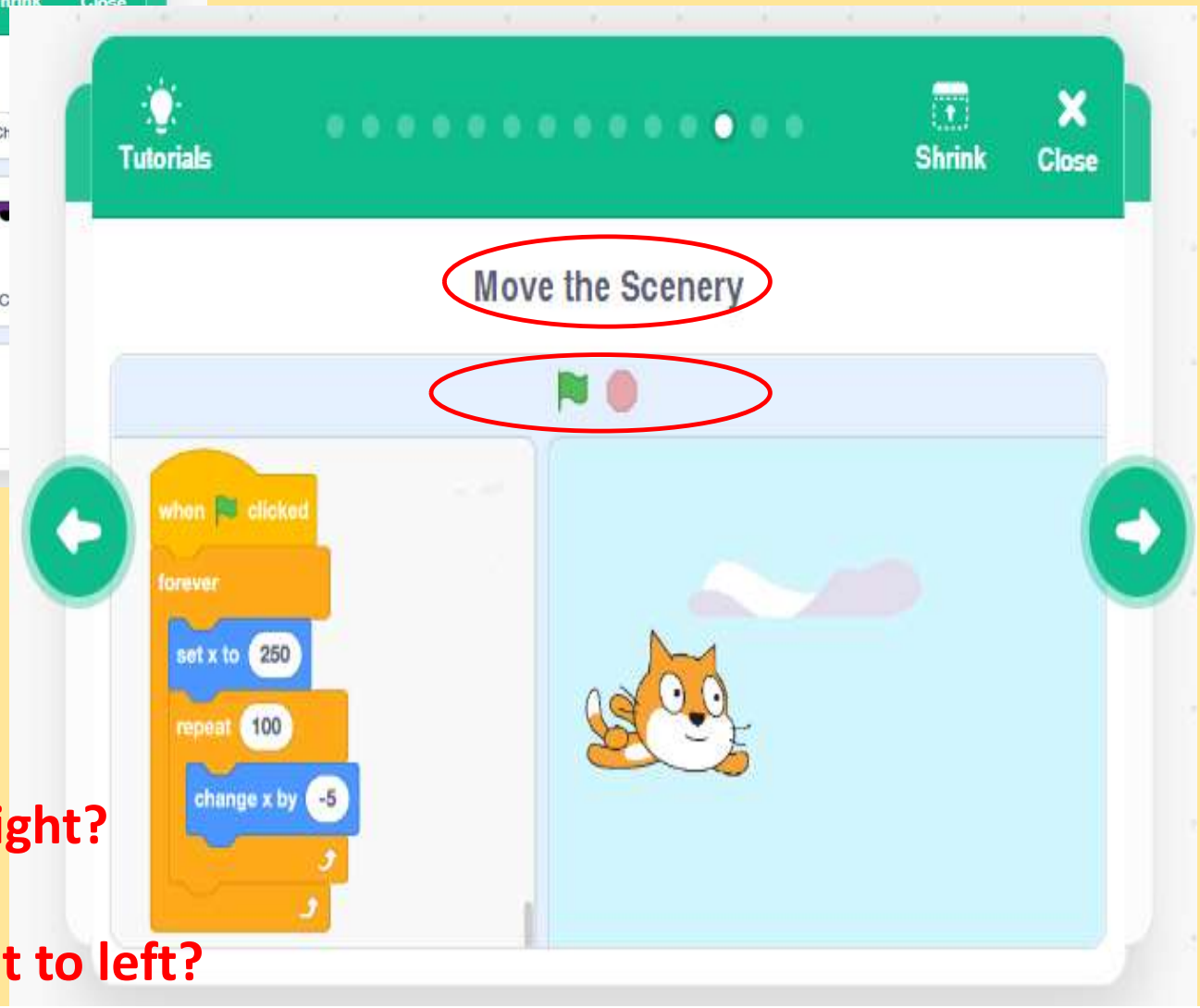


Heart moves too fast?

Software Development Life Cycle: Design, Coding, Testing, Debugging

2D X-Y: slide 66



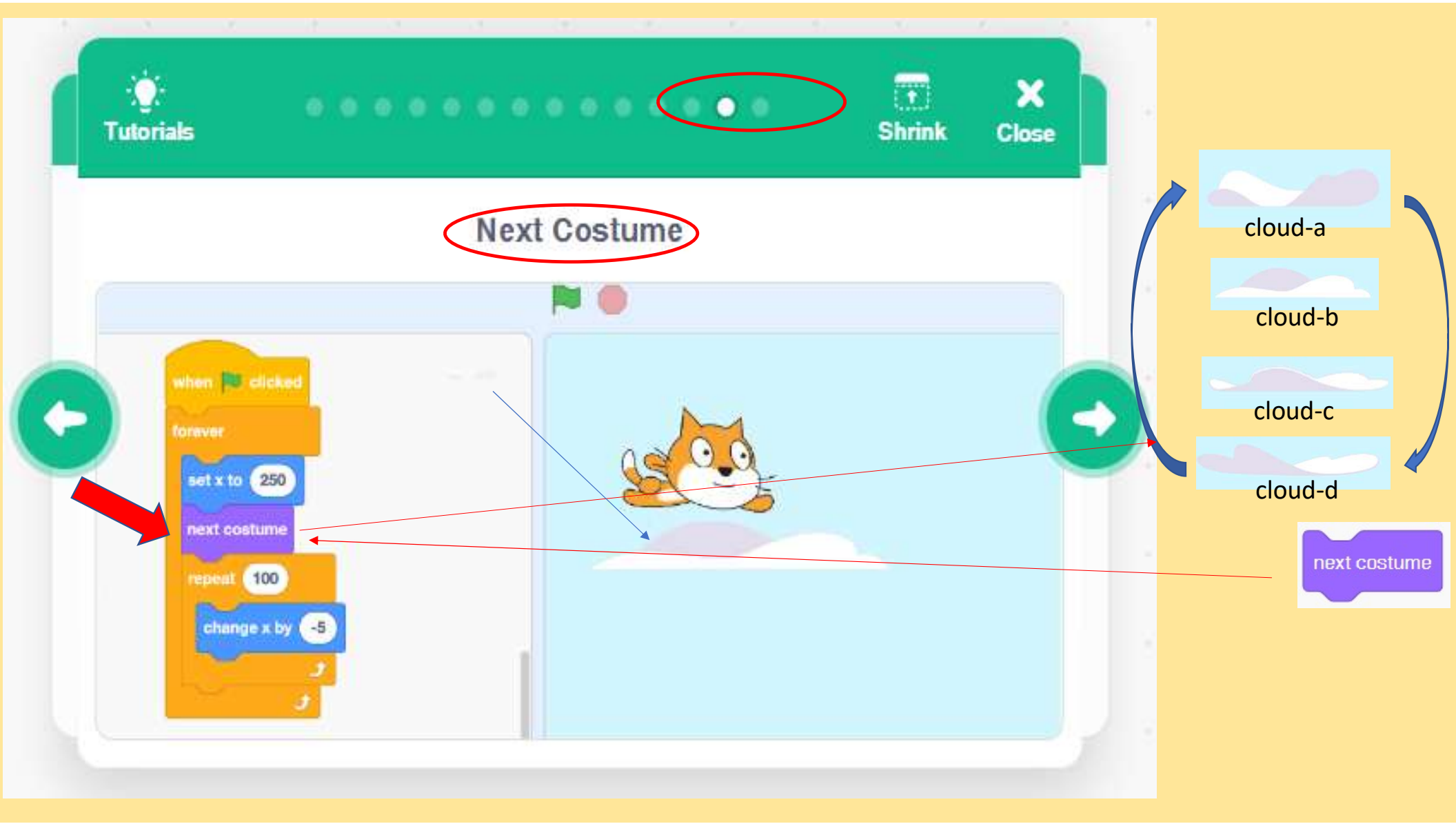


Why 250?

*** Move it to the very left or right?**

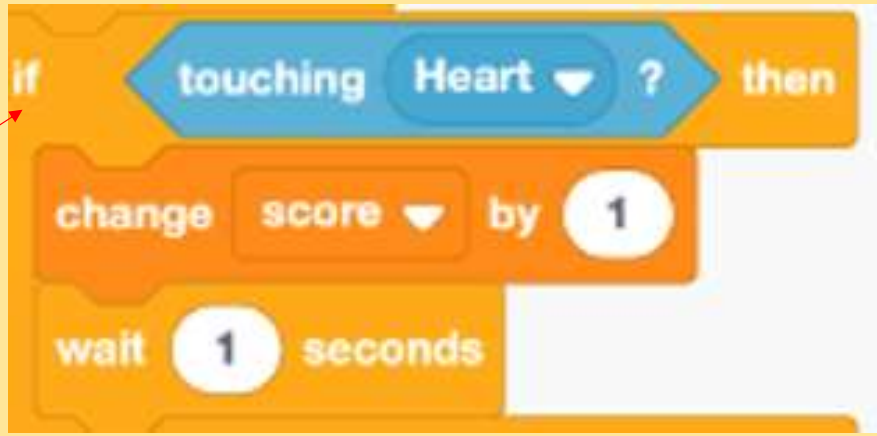
Why 100 and -5?

*** The total distance from right to left?**


Tutorials
Shrink
Close
Next Costume
when clicked
forever
set x to 250
next costume
repeat 100
change x by -5
cloud-a
cloud-b
cloud-c
cloud-d
next costume

Scratch – the basic concepts of coding

- Basic computer inputs (Camera) and outputs
- Declaring variables (Score)
- Arithmetic Operation (+, -, *, /)
- Control structures
  - Sequence
  - Repetition (Looping)
  - **Selection (Condition) if … then**

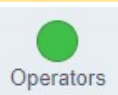


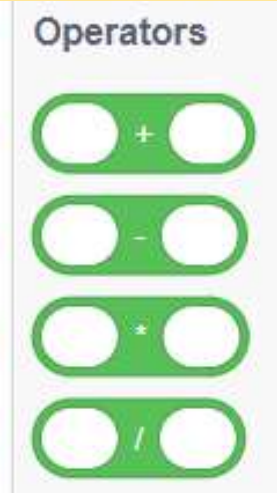
- Events (key pressed, mouse clicked, sprites overlapping, …)
- Other topics (cloning, broadcast message, ..)

- Creating **hands-on game projects**.

Challenge for you, not homework!
Conversion between temperature units (C vs. F)

- Formula: degree_C x 9/5 +32 => degree_F
  - (0°C × 9/5) + 32 = 32°F





https://www.google.com/search?q=celsius+to+fahrenheit&rlz=1C1GCEU_enUS861US861&oq=celsi&aqs=chrome.0.35i39j69i57j0i131i433i512l2j46i131i199i433i465i512j0i433i512l3j0i131i433l2.3319j0j15&sourceid=chrome&ie=UTF-8

Scratch – the basic concepts of coding

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- Arithmetic Operation (+, -, *, /)
- Control structures
  - Sequence
  - Repetition (Looping)
  - **Selection (Condition)**
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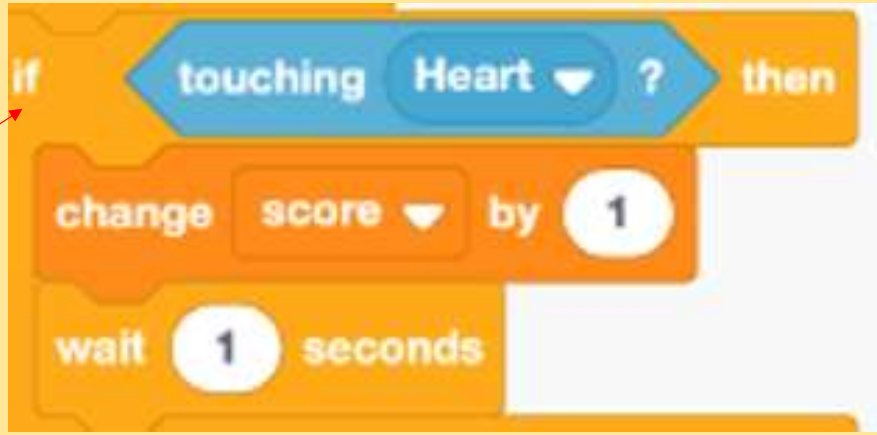
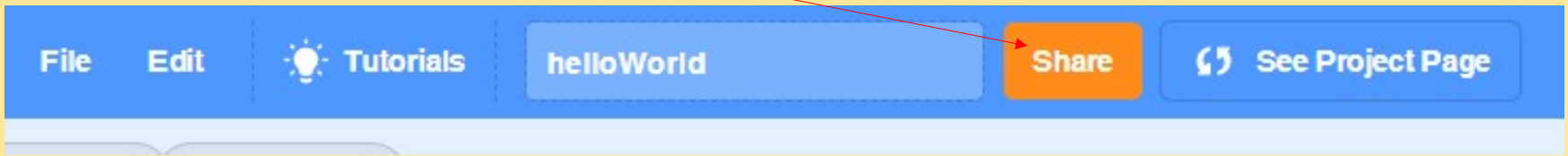


- Creating **hands-on game projects**.

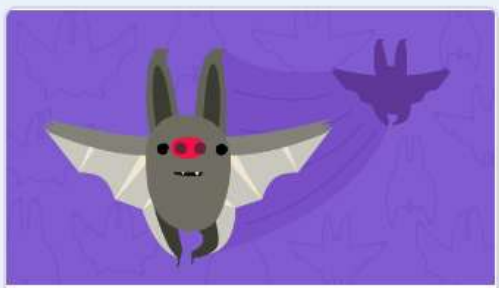
- Sharing with others !!!!!
- ReMixed



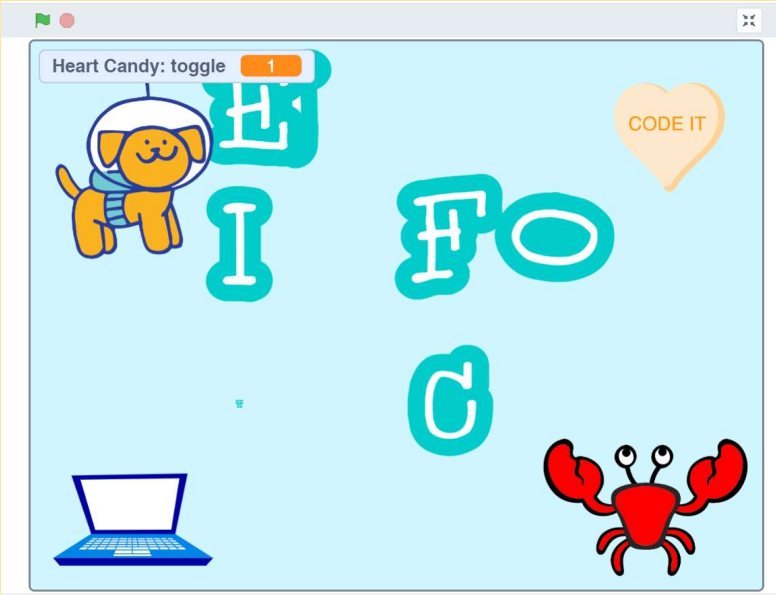
Show/Share Cases:

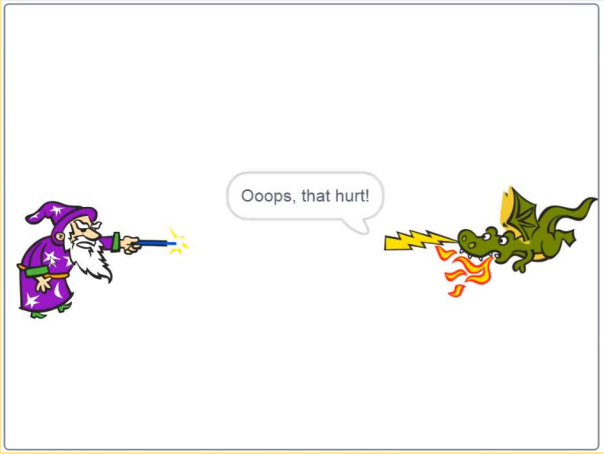
- Animate by Melissa Li

Make it Fly



Glide Around

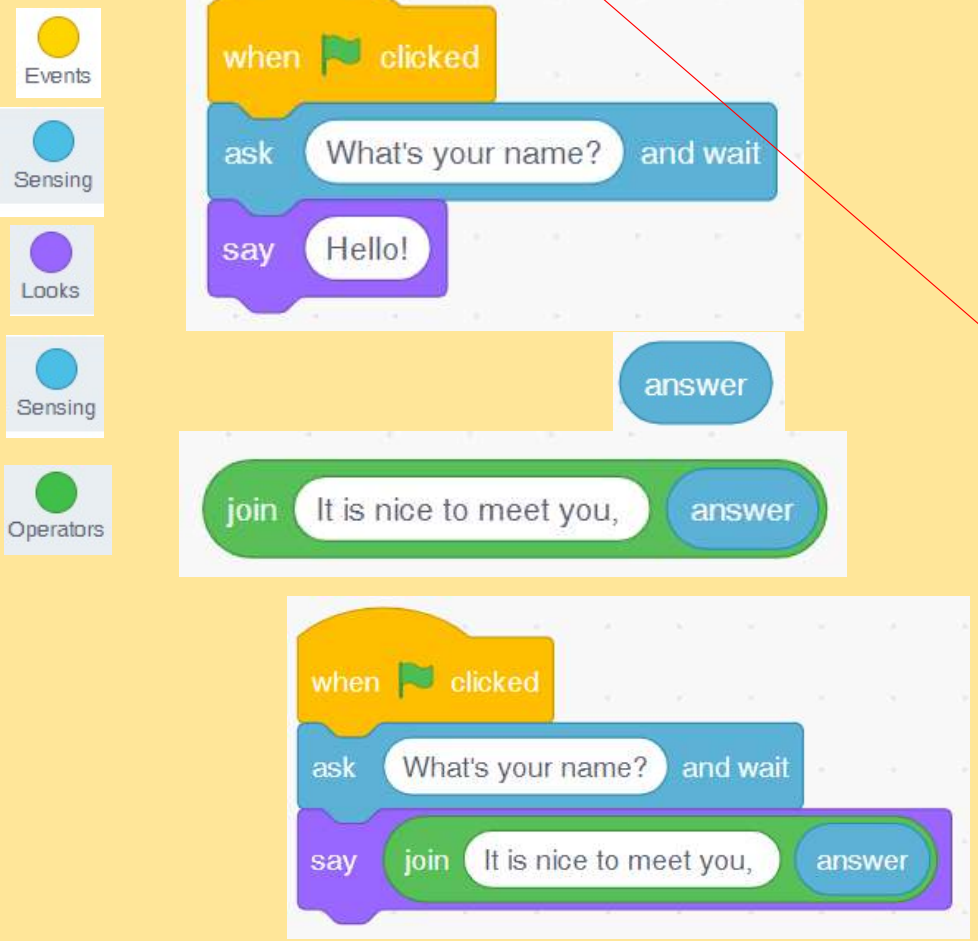


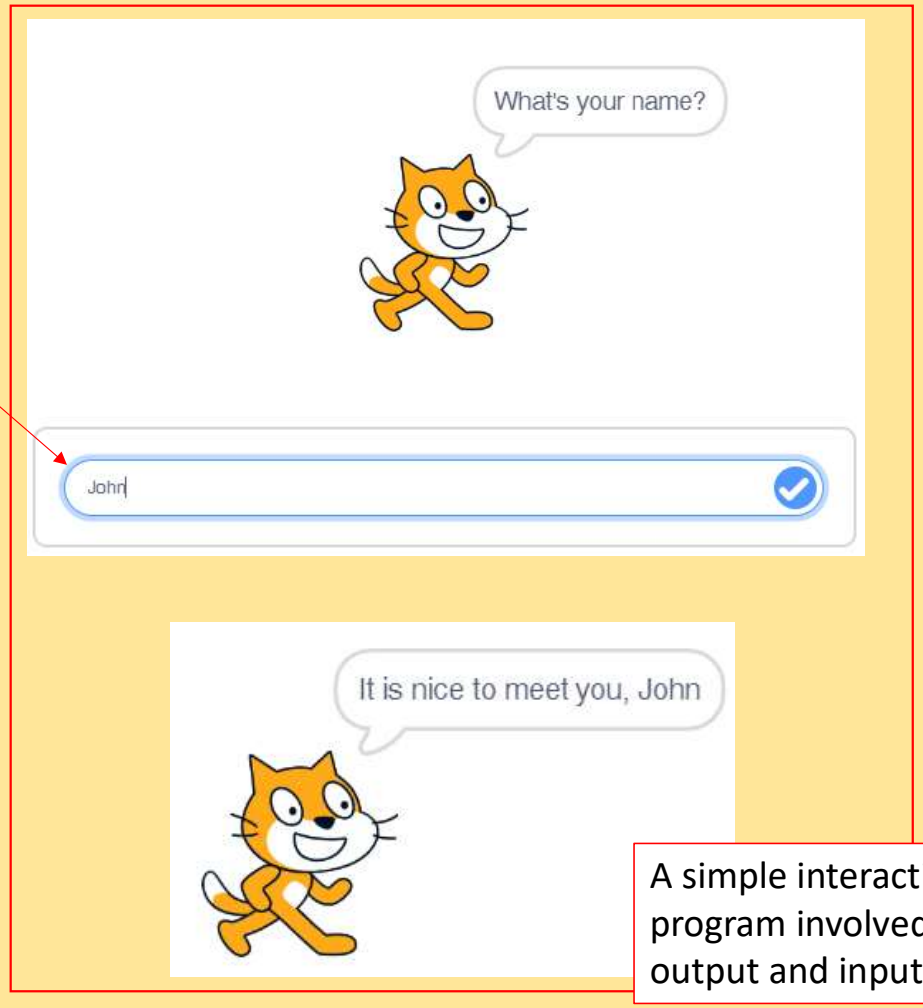


- Fun!
- Questions?

- Remaining topics

Basic Input / Output, Variable, Join (Sequencing, T-2-B)

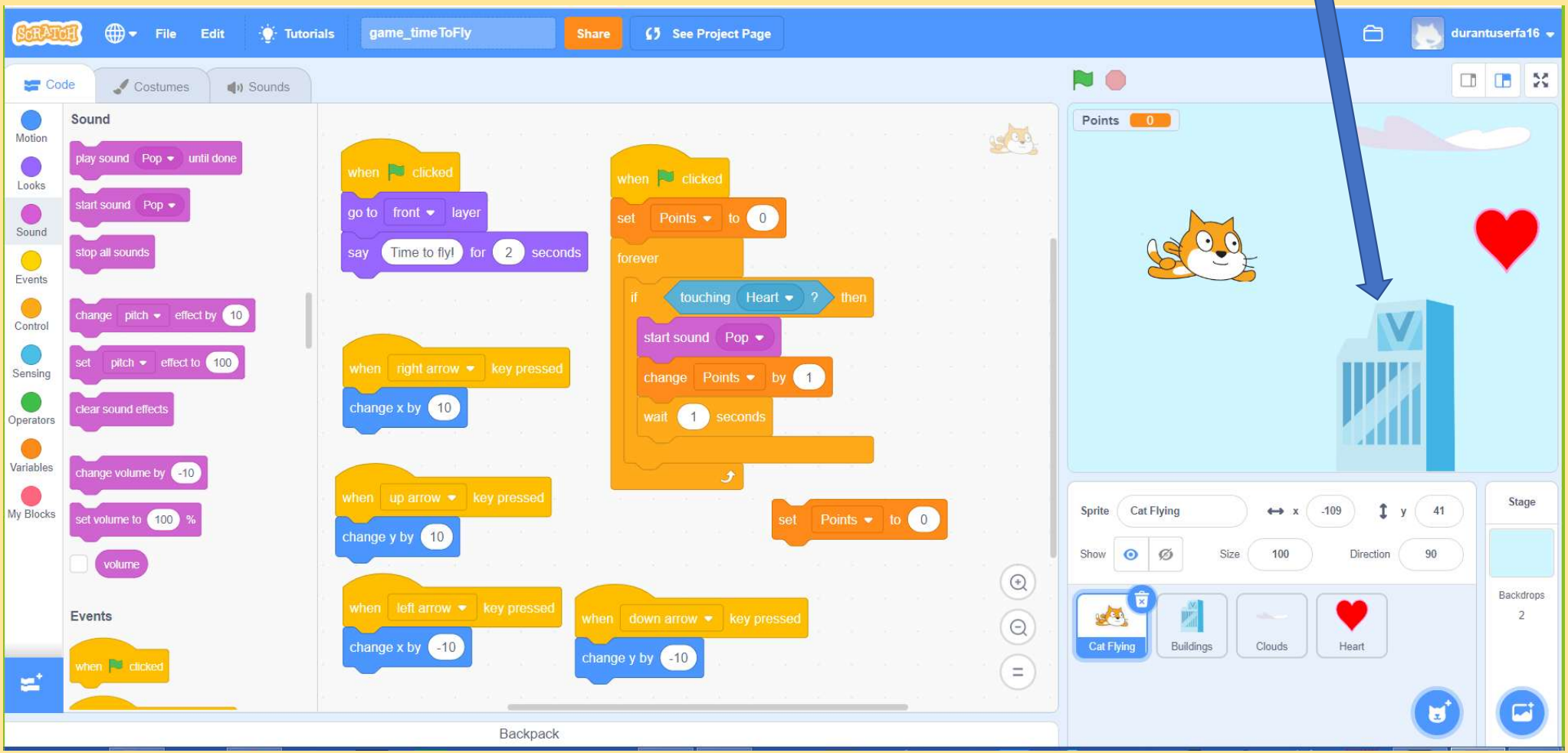




A simple interactive program involved with output and input.

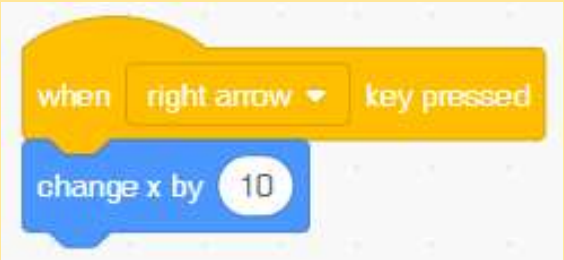
Game: Time to fly-v2!

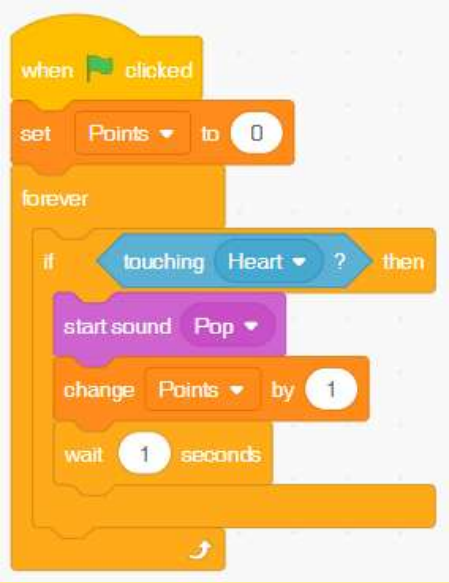
Put them all together!
(Variable, Looping, Selection, Events, Music, Building, …)

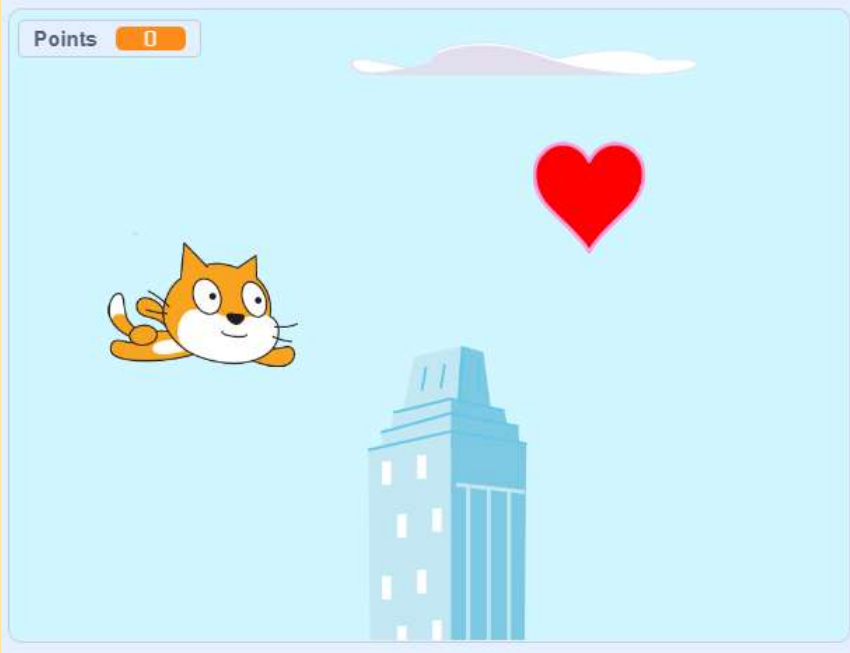


https://youtu.be/uBTFqUr7KPA

Events (key pressed, mouse clicked, …):
Move the Sprite in all directions using arrow keys







https://youtu.be/uBTFqUr7KPA

Quick show on the fly-cards.pdf

Make it Fly Cards

Use these cards in this order:

1. Choose a Character
2. Start Flying
3. Switch Looks
4. Make it Interactive
5. Floating Clouds
6. Flying Hearts
7. Collect Points

Choose a Character

Choose a character to fly.

Choose a Character

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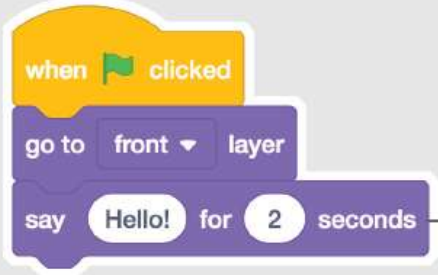
GET READY

Choose a backdrop.

Choose a sprite from the **Flying** theme.

ADD THIS CODE



Type what you want your sprite to say.

TRY IT

Click the green flag to start

Start Flying

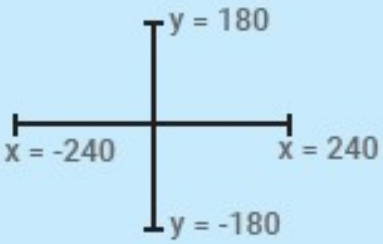
scratch.mit.edu

GET READY

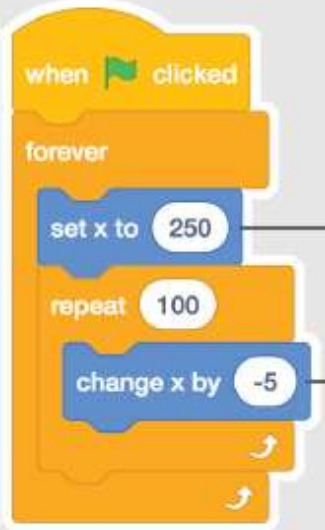
Choose a sprite to fly by, such as Buildings.

TIP

x is the position on the Stage from left to right.



ADD THIS CODE



Start from the right end of the stage.

Type a negative number to move left.

TRY IT

Click the green flag to start

Switch Looks

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GET READY

Click to select the **Buildings** sprite.

Then, click the **Costumes** tab to see different costumes.

ADD THIS CODE

Click the Code tab.

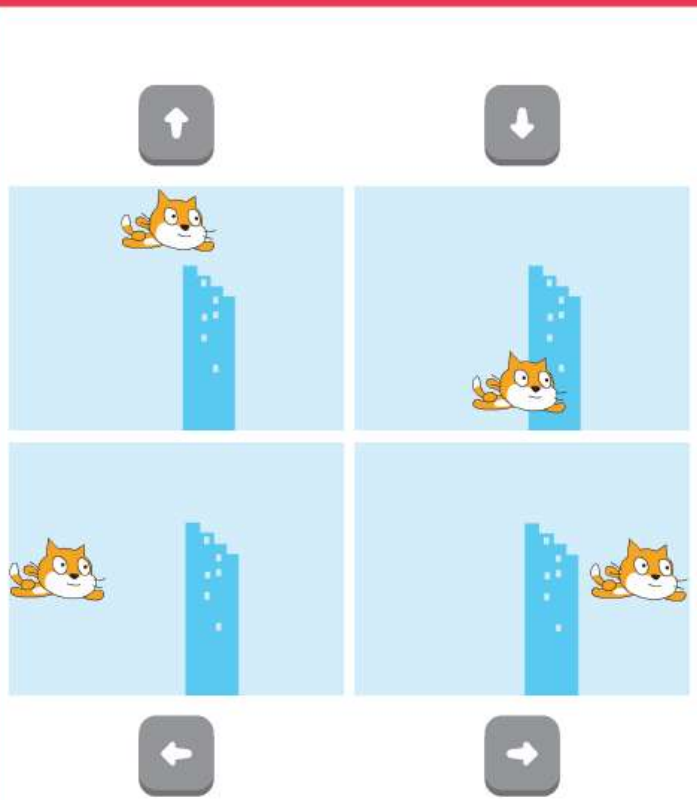
Add this block to switch costumes.

TRY IT

Click the green flag to start

Make It Interactive

Make your character move
when you press a key.



Choose a Character

Choose a character to fly.

Choose a Character

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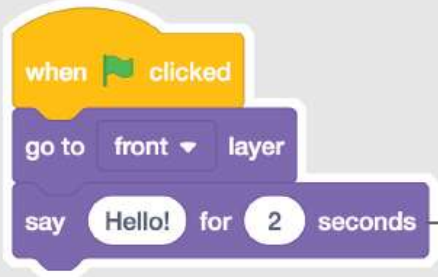
GET READY

Choose a backdrop.

Choose a sprite from the **Flying** theme.

ADD THIS CODE



Type what you want your sprite to say.

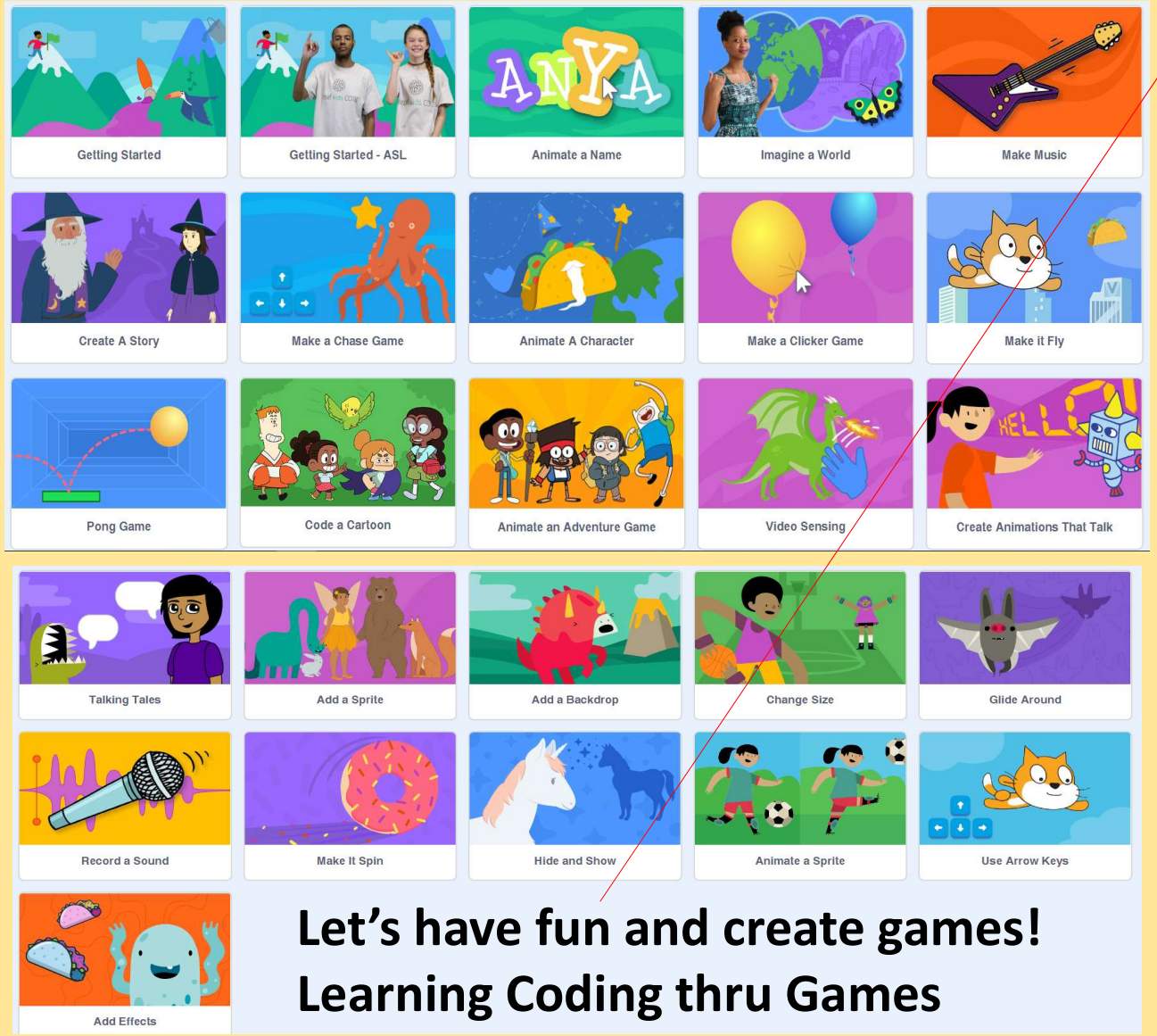
Make it Fly Cards

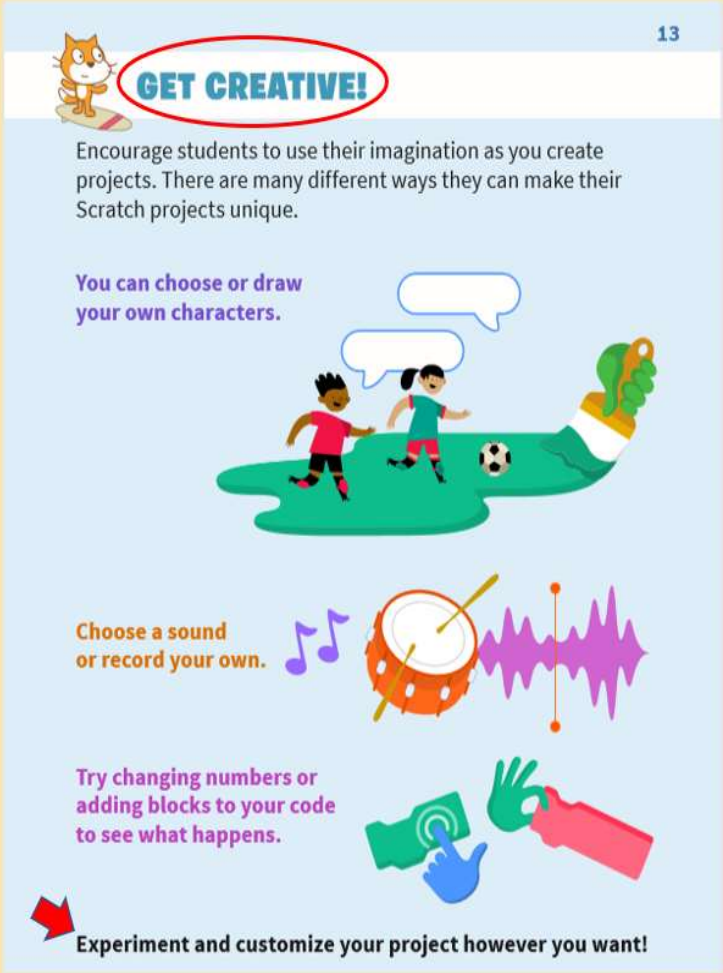
Choose any character and make it fly!

Events (key pressed, mouse clicked, …):
Move the Sprite in all directions using arrow keys

- Refer to fly-cards.pdf for more information.
- What is next?

Can you design the Pong Game for your family?



Let's have fun and create games!
Learning Coding thru Games

Or STEM (Science Technology Engineering Mathematics) - Projects